جامعة الملك عبدالله
للعلوم والتقنية
King Abdullah University of
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CS229 Machine Learning

Introduction

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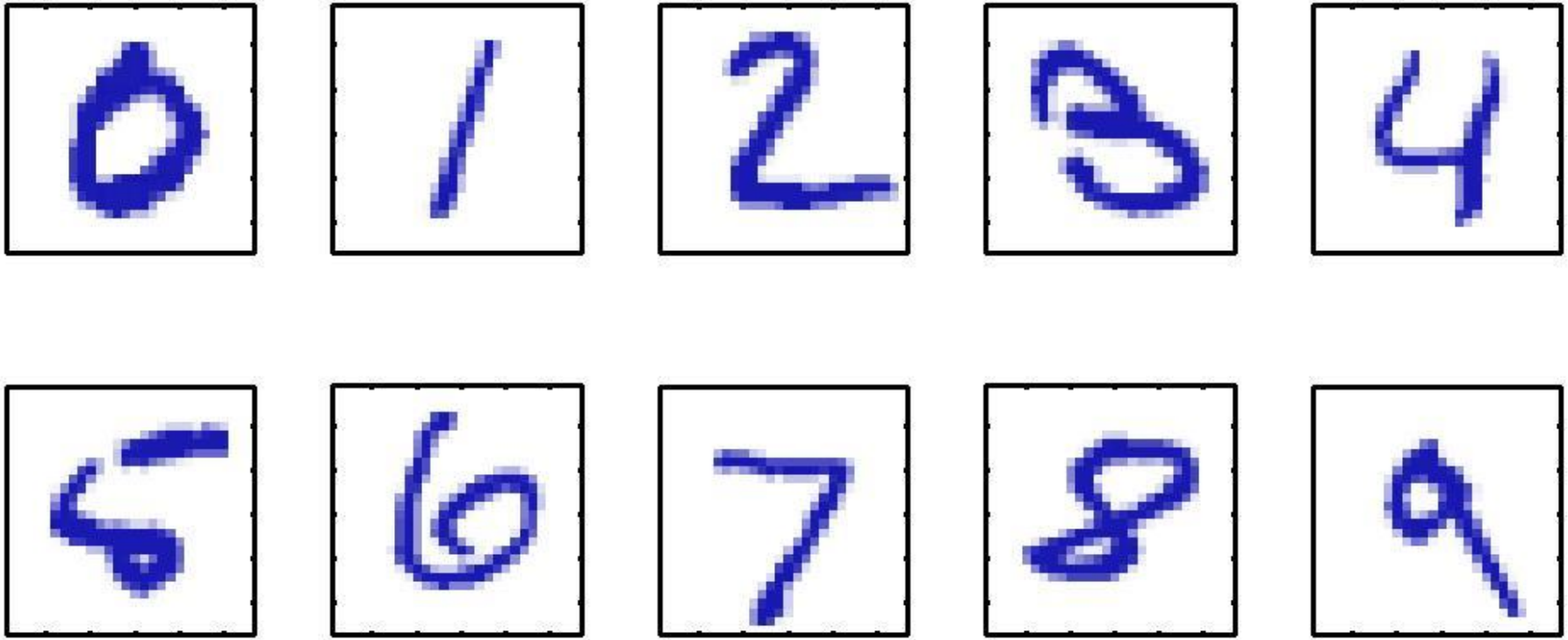
Handwriting Recognition



Basic Machine Learning Terminology

- *Input vector* $\boldsymbol{x}$ one image of 784 (28×28) floats
- *Training set* $\{\boldsymbol{x}_1 \cdots \boldsymbol{x}_N\}$ of N images
- *Target vector* $\boldsymbol{t}$ determines category of image $\boldsymbol{x}$
- In *training (learning) phase* machine learns function $\boldsymbol{y}(\boldsymbol{x})$
- Output of $\boldsymbol{y}$ has the format as the target $\boldsymbol{t}$
- *Generalization* characterizes ability to correctly categorize new images (from *test set*)
- Data is *preprocessed* into a uniform format

Basic Machine Learning Terminology

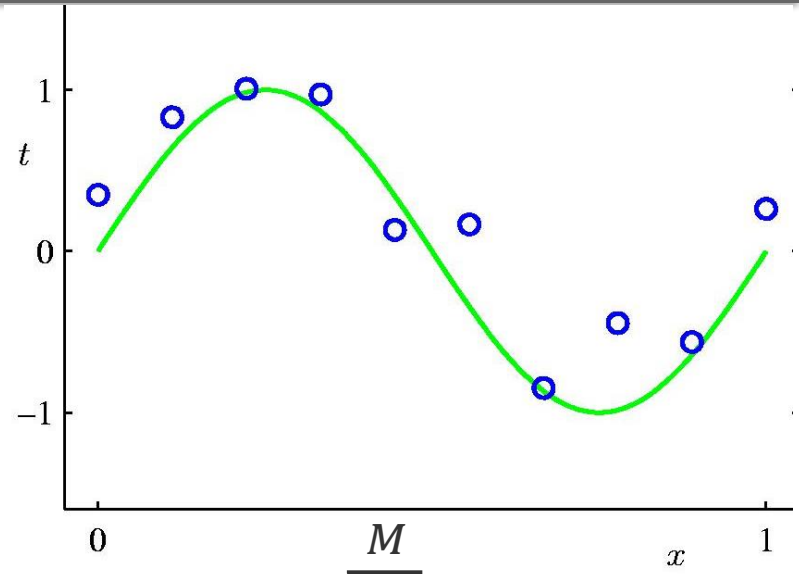
- *Supervised learning* using training data associated with the corresponding target values
- *Classification* is the assignment of each input vector into a discrete category
- *Regression* is the estimation of a continuous variable
- *Unsupervised learning* does not have target values associated with a training set
- Discovers distribution patterns in the data through *clustering* or *density estimation*

Basic Machine Learning Terminology

- *Reinforcement learning* estimates actions that lead to the highest reward
- Single step is not associated with optimal output
- Longer sequence of steps leads to an outcome
- *Credit assigns* reward to individual steps
- *Exploration* and *exploitation* are two patterns

Polynomial Curve Fitting

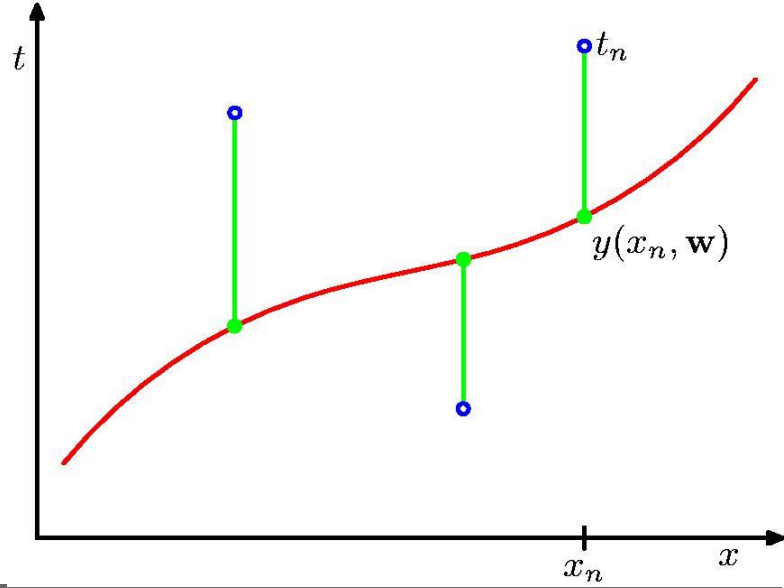
- Input x associated with target t
- Synthetics data generated by $\sin(2\pi x)$ with added Gaussian noise
- Training set $\mathbf{x} \equiv (x_1 \cdots x_N)^{\mathrm{T}}$
- Observations $\mathbf{t} \equiv (t_1 \cdots t_N)^{\mathrm{T}}$
- Goal: Make predictions $\hat{t}$ for new $\hat{x}$
- Linear model:



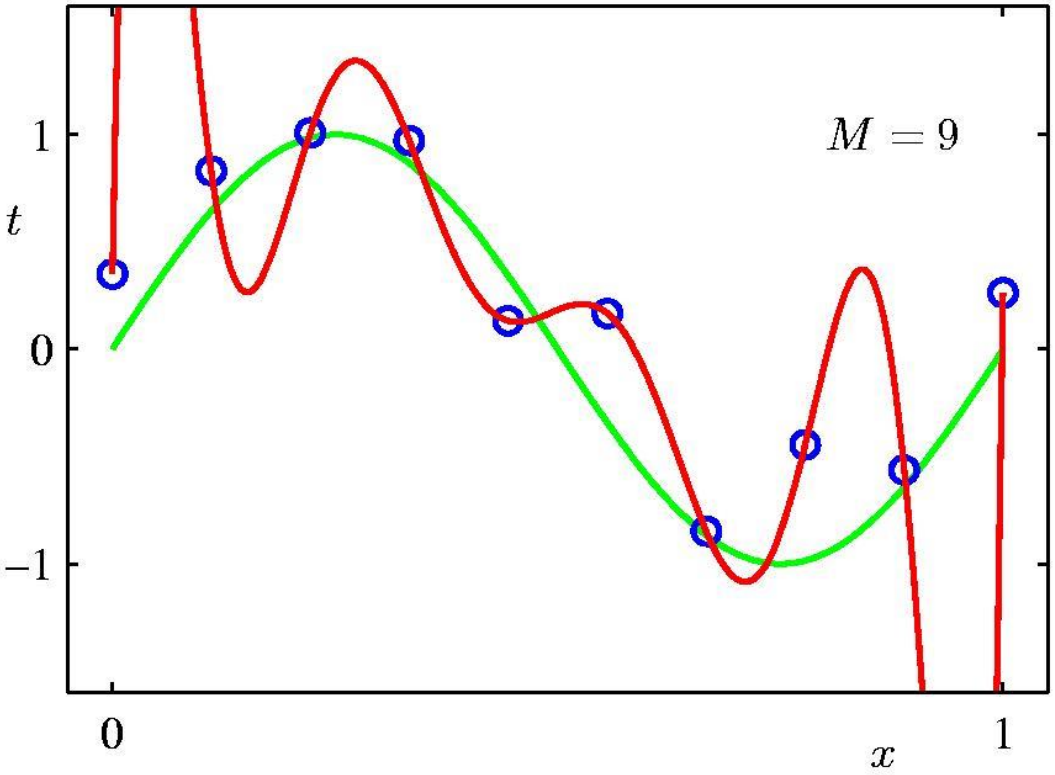
$$y(x, \boldsymbol{w}) = w_0 x^0 + w_1 x^1 + w_2 x^2 + \cdots + w_M x^M = \sum_{j=0}^{M} w_j x^j$$

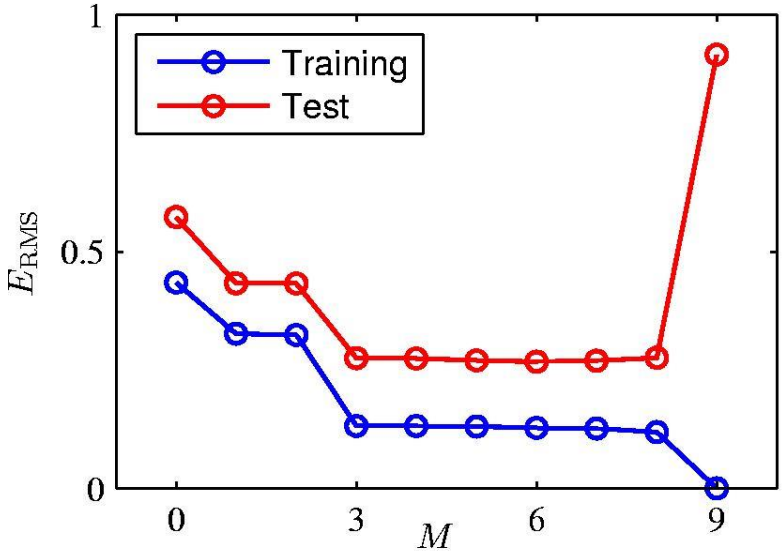
Polynomial Curve Fitting

- Approach: Find best value of the coefficients $\boldsymbol{w}^*$ through minimizing a sum-of-squares *error function*
- $E(\boldsymbol{w}) = \frac{1}{2}\sum_{n=1}^{N}\{y(x_n, \boldsymbol{w}) - t_n\}^2$
- *Model selection* determines choice of polynomial degree
- $E_{RMS} = \sqrt{2E(\boldsymbol{w}^*)/N}$



Polynomial Curve Fitting





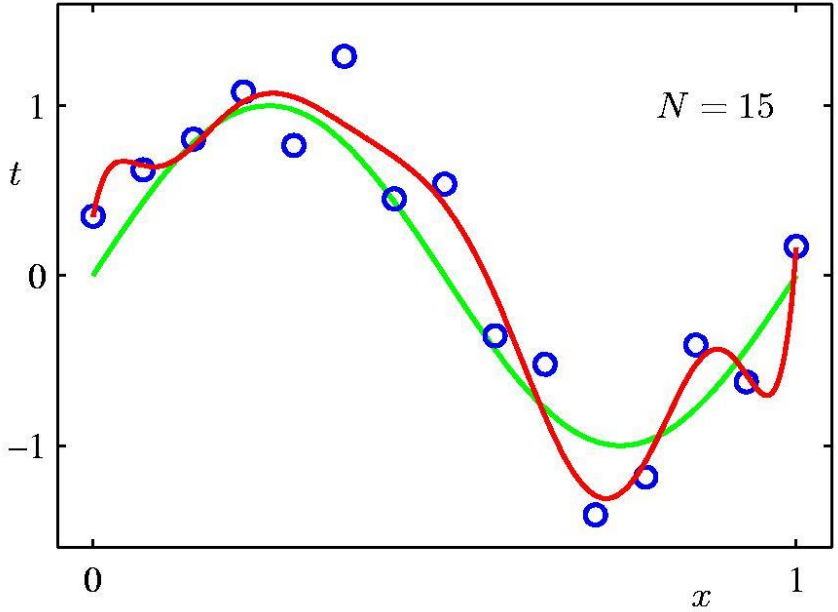
Polynomial Curve Fitting

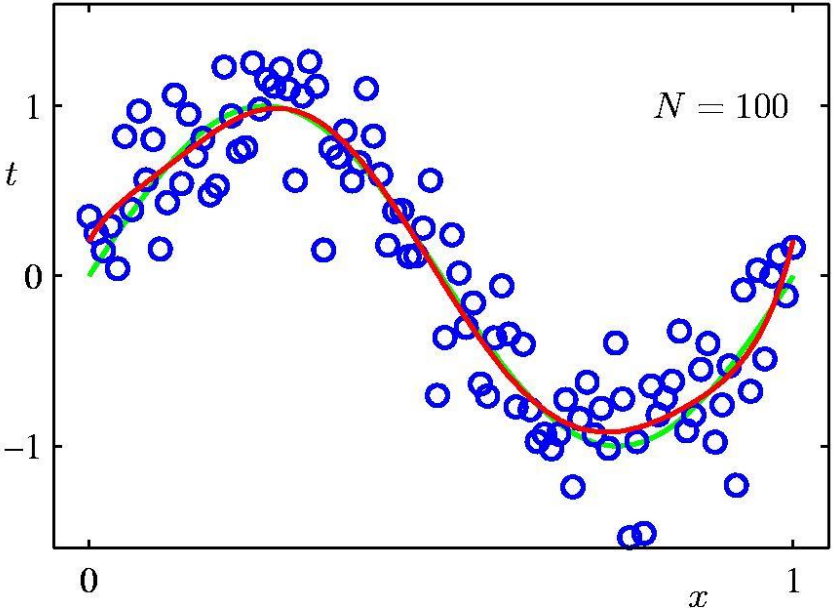
- Increasing M increases abs value of $\boldsymbol{w}^*$ coefficients
- Large $\boldsymbol{w}^*$ coefficients are associated with oscillations that are fitted to random noise
- *Overfitting* leads to perfect performance on training set and poor performance on test set

	$M=0$	$M=1$	$M=3$	$M=9$
$w_0^\star$	0.19	0.82	0.31	0.35
$w_1^\star$		-1.27	7.99	232.37
$w_2^\star$			-25.43	-5321.83
$w_3^\star$			17.37	48568.31
$w_4^\star$				-231639.30
$w_5^\star$				640042.26
$w_6^\star$				-1061800.52
$w_7^\star$				1042400.18
$w_8^\star$				-557682.99
$w_9^\star$				125201.43

Polynomial Curve Fitting

- More data reduces the effect of overfitting





Polynomial Curve Fitting

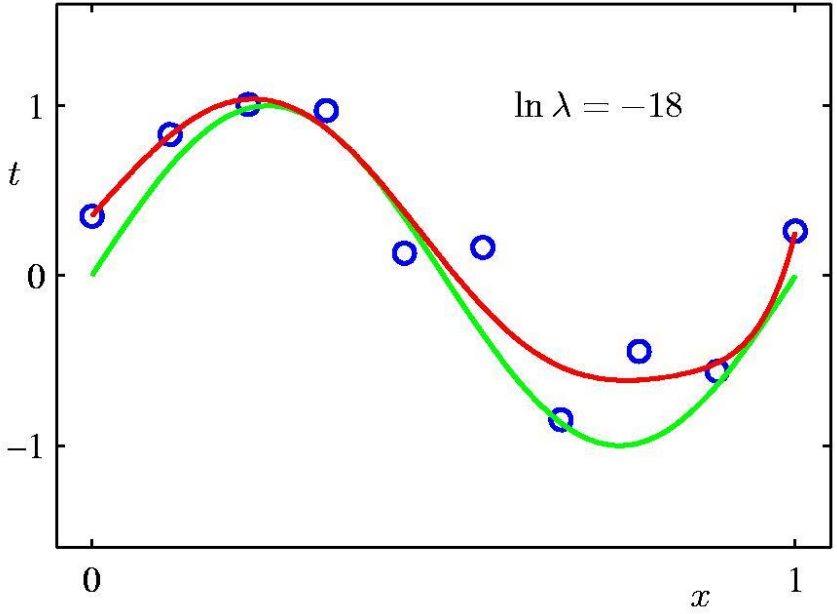
- *Regularization* adds a penalty term to the error function to discourage large absolute value of $\boldsymbol{w}^*$ coefficients
- *Shrinkage method* known as *ridge regression* or *weight decay*

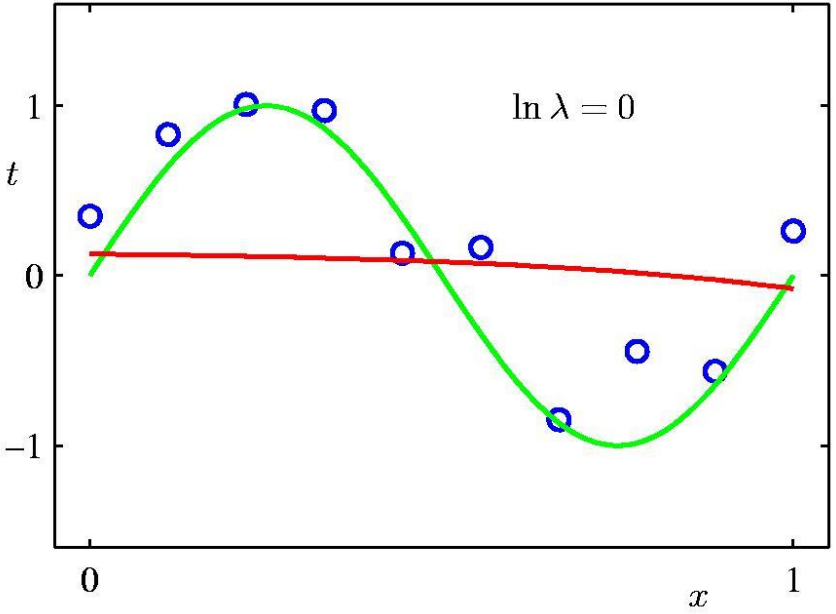
$$\tilde{E}(\boldsymbol{w}) = \frac{1}{2}\sum_{n=1}^{N}\{y(x_n, \boldsymbol{w}) - t_n\}^2 + \frac{\lambda}{2}\|\boldsymbol{w}^2\|$$

$$\|\boldsymbol{w}^2\| \equiv \boldsymbol{w}^T\boldsymbol{w} = w_0^2 + w_1^2 + w_2^2 + \cdots + w_M^2$$

Polynomial Curve Fitting

- Value λ controls effective complexity of the model

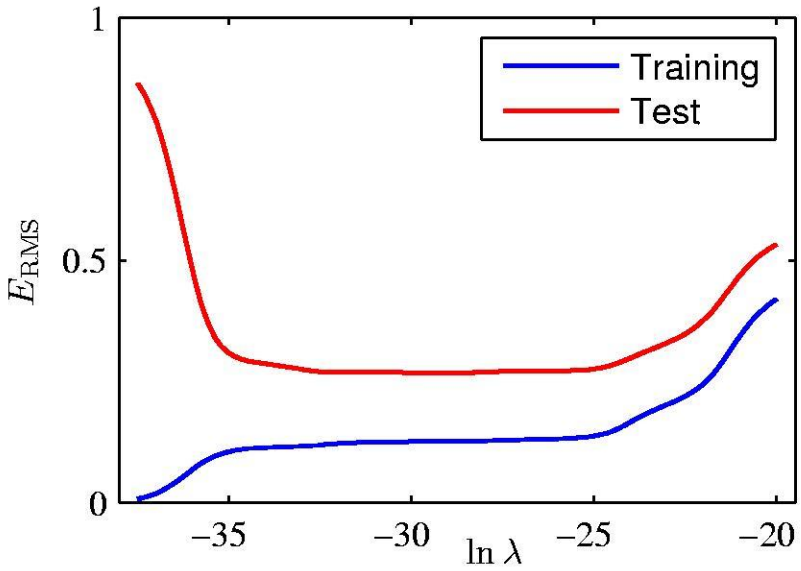




Polynomial Curve Fitting

- Value λ controls effective complexity of the model

	$\ln \lambda = -\infty$	$\ln \lambda = -18$	$\ln \lambda = 0$
$w_0^\star$	0.35	0.35	0.13
$w_1^\star$	232.37	4.74	-0.05
$w_2^\star$	-5321.83	-0.77	-0.06
$w_3^\star$	48568.31	-31.97	-0.05
$w_4^\star$	-231639.30	-3.89	-0.03
$w_5^\star$	640042.26	55.28	-0.02
$w_6^\star$	-1061800.52	41.32	-0.01
$w_7^\star$	1042400.18	-45.95	-0.00
$w_8^\star$	-557682.99	-91.53	0.00
$w_9^\star$	125201.43	72.68	0.01



Closed Form Linear Regression

$$2E(\boldsymbol{w}) = \sum_{n=1}^{N} \{y(x_n, \boldsymbol{w}) - t_n\}^2 = \sum_{n=1}^{N} \{w_0 + w_1 x_n - t_n\}^2$$

$$\frac{\partial 2E(\boldsymbol{w}^*)}{\partial w_0} = \sum_{n=1}^{N} 2\{w_0^* + w_1^* x_n - t_n\} = 0$$

$$\Rightarrow N w_0^* = \sum_{n=1}^{N} t_n - w_1^* \sum_{n=1}^{N} x_n \Rightarrow w_0^* = \bar{t} - w_1^* \bar{x}$$

Closed Form Linear Regression

$$\frac{\partial 2E(\boldsymbol{w}^*)}{\partial w_1} = \sum_{n=1}^{N} 2\{w_0^* + w_1^* x_n - t_n\}x_n = 2(\bar{t} - w_1^*\bar{x})\sum_{n=1}^{N} x_n + 2w_1^*\sum_{n=1}^{N} x_n^2 - 2\sum_{n=1}^{N} x_n t_n$$

$$\Rightarrow w_1^*\left(\sum_{n=1}^{N} x_n^2 - \bar{x}\sum_{n=1}^{N} x_n\right) = \sum_{n=1}^{N} x_n t_n - \bar{t}\sum_{n=1}^{N} x_n = w_1^*\left(\sum_{n=1}^{N} x_n^2 - N\bar{x}\bar{x}\right) = \sum_{n=1}^{N} x_n t_n - N\bar{x}\bar{t}$$

$$= w_1^*\left(\sum_{n=1}^{N}(x_n - \bar{x})^2\right) = \sum_{n=1}^{N}(x_n - \bar{x})(t_n - \bar{t}) \Rightarrow w_1^* = \frac{\sum_{n=1}^{N}(x_n - \bar{x})(t_n - \bar{t})}{\sum_{n=1}^{N}(x_n - \bar{x})^2}$$

$$\sum_{n=1}^{N}(x_n - \bar{x})^2 = \sum_{n=1}^{N}(x_n^2 - x_n\bar{x} - \bar{x}x_n + \bar{x}\bar{x}) = \sum_{n=1}^{N} x_n^2 - \bar{x}\sum_{n=1}^{N} x_n - \bar{x}\sum_{n=1}^{N} x_n + N\bar{x}\bar{x}$$

$$= \sum_{n=1}^{N} x_n^2 - N\bar{x}\bar{x} - N\bar{x}\bar{x} + N\bar{x}\bar{x} = \sum_{n=1}^{N} x_n^2 - N\bar{x}\bar{x}$$

Closed Form Linear Regression

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{w} \quad \boldsymbol{X} \in \mathbb{R}^{N\times d} \quad \boldsymbol{w} \in \mathbb{R}^{d} \quad \boldsymbol{t} \in \mathbb{R}^{N}$$

$$2E(\boldsymbol{w}) = (\boldsymbol{y} - \boldsymbol{t})^2 = (\boldsymbol{X}\boldsymbol{w} - \boldsymbol{t})^T(\boldsymbol{X}\boldsymbol{w} - \boldsymbol{t})$$

$$= (\boldsymbol{X}\boldsymbol{w})^T\boldsymbol{X}\boldsymbol{w} - 2\boldsymbol{t}^T\boldsymbol{X}\boldsymbol{w} + \boldsymbol{t}^T\boldsymbol{t} = \boldsymbol{w}^T\boldsymbol{X}^T\boldsymbol{X}\boldsymbol{w} - 2(\boldsymbol{X}^T\boldsymbol{t})^T\boldsymbol{w} + \boldsymbol{t}^T\boldsymbol{t}$$

$$\nabla_{\boldsymbol{w}} E(\boldsymbol{w}^*) = \frac{1}{2}(2\boldsymbol{X}^T\boldsymbol{X}\boldsymbol{w}^* - 2\boldsymbol{X}^T\boldsymbol{t}) = \boldsymbol{0}$$

$$\Rightarrow \boldsymbol{X}^T\boldsymbol{X}\boldsymbol{w}^* = \boldsymbol{X}^T\boldsymbol{t} \Rightarrow \boldsymbol{w}^* = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{t}$$

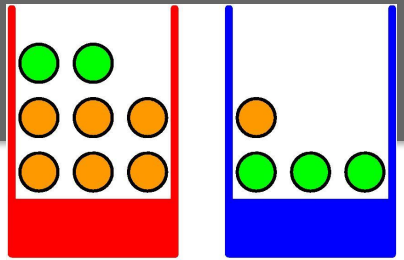
$\nabla_{\boldsymbol{w}}[\boldsymbol{w}^T\boldsymbol{A}\boldsymbol{w}] = (\boldsymbol{A} + \boldsymbol{A}^T)\boldsymbol{w} = 2\boldsymbol{A}\boldsymbol{w}$ (Matrix $\boldsymbol{A}$ is symmetric).

Closed Form Linear Regression with Regularization

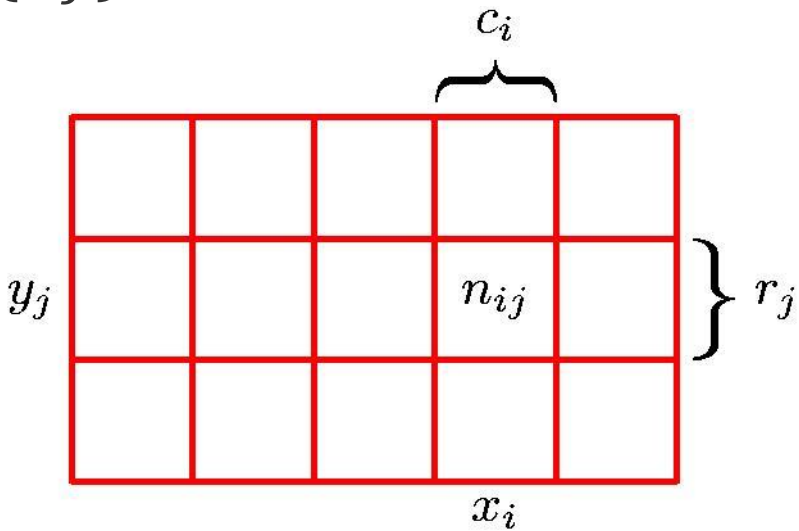
$$2\tilde{E}(\boldsymbol{w}) = (\boldsymbol{y} - \boldsymbol{t})^2 + \lambda(\boldsymbol{w}^2)$$
$$= (\boldsymbol{X}\boldsymbol{w} - \boldsymbol{t})^T(\boldsymbol{X}\boldsymbol{w} - \boldsymbol{t}) + \lambda \boldsymbol{w}^T\boldsymbol{w}$$
$$= \boldsymbol{w}^T\boldsymbol{X}^T\boldsymbol{X}\boldsymbol{w} - 2(\boldsymbol{X}^T\boldsymbol{t})^T\boldsymbol{w} + \boldsymbol{t}^T\boldsymbol{t} + \lambda \boldsymbol{w}^T\boldsymbol{w}$$

$$\nabla_{\boldsymbol{w}}\tilde{E}(\boldsymbol{w}^*) = \frac{1}{2}(2\boldsymbol{X}^T\boldsymbol{X}\boldsymbol{w}^* - 2\boldsymbol{X}^T\boldsymbol{t} + 2\lambda\boldsymbol{w}^*) = \boldsymbol{0}$$
$$\Rightarrow (\boldsymbol{X}^T\boldsymbol{X} + \lambda\boldsymbol{I})\boldsymbol{w}^* = \boldsymbol{X}^T\boldsymbol{t}$$
$$\Rightarrow \boldsymbol{w}^* = (\boldsymbol{X}^T\boldsymbol{X} + \lambda\boldsymbol{I})^{-1}\boldsymbol{X}^T\boldsymbol{t}$$

Probability Theory Review



- Random variables X and Y
- $X = \{x_i\}: i = 1, \dots, M, Y = \{y_j\}, j = 1, \dots, L$
- Total number of trials N
- n_{ij} is number of trials where $X = x_i, Y = y_j$



Probability Theory Review

- Joint probability:

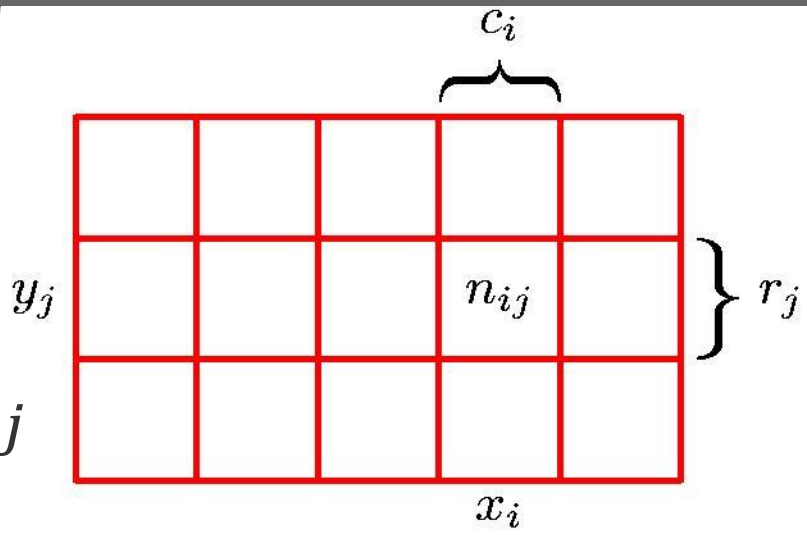
$$p(X = x_i, Y = y_j) = \frac{n_{ij}}{N}$$

- Marginal probability:

$$p(X = x_i) = \frac{c_i}{N}, c_i = \sum_{j=1}^{L} n_{ij}$$

- Sum rule:

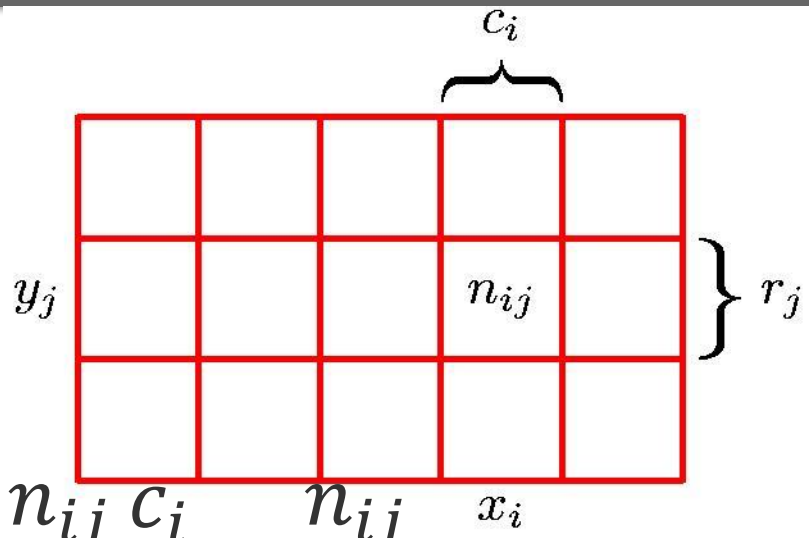
$$p(X = x_i) = \sum_{j=1}^{L} p(X = x_i, Y = y_j)$$



Probability Theory Review

- Conditional probability:

$$p(Y = y_j | X = x_i) = \frac{n_{ij}}{c_i}$$



- Product rule:

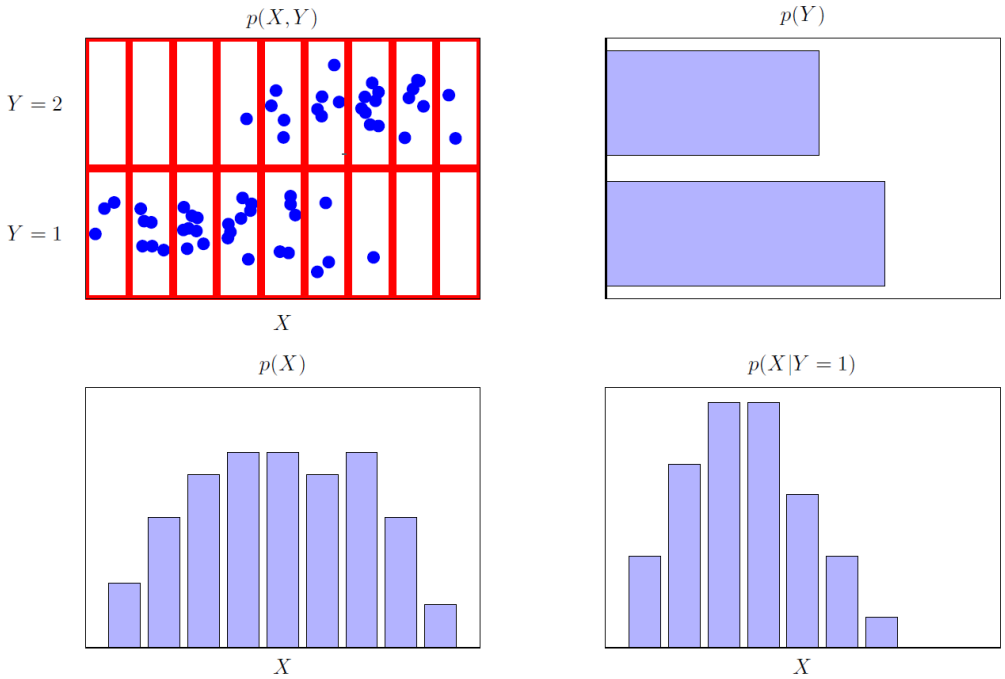
$$p(X = x_i, Y = y_j) = \frac{n_{ij}}{c_i}\frac{c_i}{N} = \frac{n_{ij}}{N}$$

$$= p(Y = y_j | X = x_i)p(X = x_i)$$

Probability Theory Review

- Bayes theorem:

$$p(Y|X) = \frac{p(X|Y)p(Y)}{p(X)}$$



Probability Theory Review

- Box $B = \{r, b\}$ fruit $F = \{a, o\}$

$$p(B = r) = \frac{4}{10}, p(B = b) = \frac{6}{10}$$

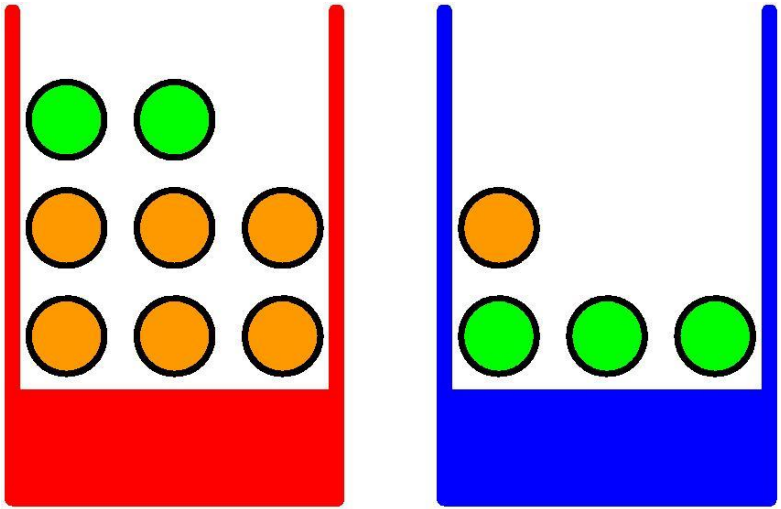
$$p(B = r) + p(B = b) = 1$$

$$p(F = a|B = r) = \frac{1}{4}$$

$$p(F = o|B = r) = \frac{3}{4}$$

$$p(F = a|B = b) = \frac{3}{4}$$

$$p(F = o|B = b) = \frac{1}{4}$$



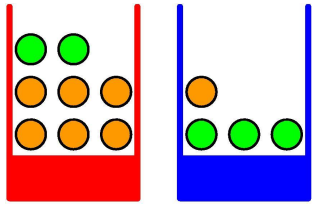
Probability Theory Review

$$p(F=a) = p(F=a|B=r)p(B=r) + p(F=a|B=b)p(B=b)$$

$$= \frac{1}{4}\frac{4}{10} + \frac{3}{4}\frac{6}{10} = \frac{11}{20}$$

$$p(F=o) = 1 - \frac{11}{20} = \frac{9}{20}$$

$$p(B=r|F=o) = \frac{p(F=o|B=r)p(B=r)}{p(F=o)} = \frac{\frac{3}{4}\frac{4}{10}}{\frac{9}{20}} = \frac{2}{3}$$



Probability Theory Review

- *Prior probability* $p(X)$ before an event happen
- *Posterior probability* $p(X|Y)$ after an event happen that determined value of Y
- If $p(X,Y) = p(X)p(Y)$ then the two distributions are *independent* and $p(Y|X) = p(Y)$

Probability Theory Review

- *Continuous random variables*

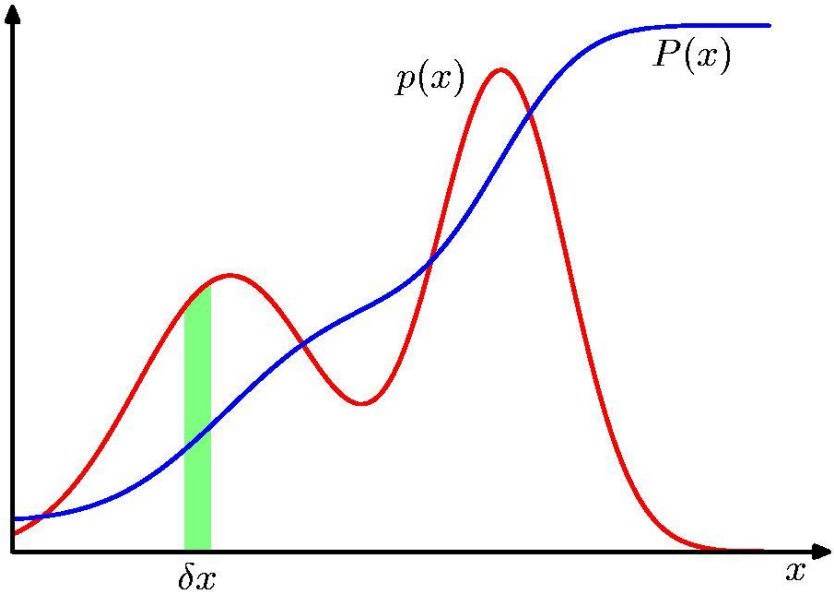
$$p\big(x \in (a,b)\big) = \int_a^b p(x)dx$$

$$P(z) = \int_{-\infty}^{z} p(x)dx$$

$$\int_{-\infty}^{\infty} p(x)dx = 1$$

$$p(x) = \int p(x,y)dy$$

$$p(x,y) = p(y|x)p(x)$$



Probability Theory Review

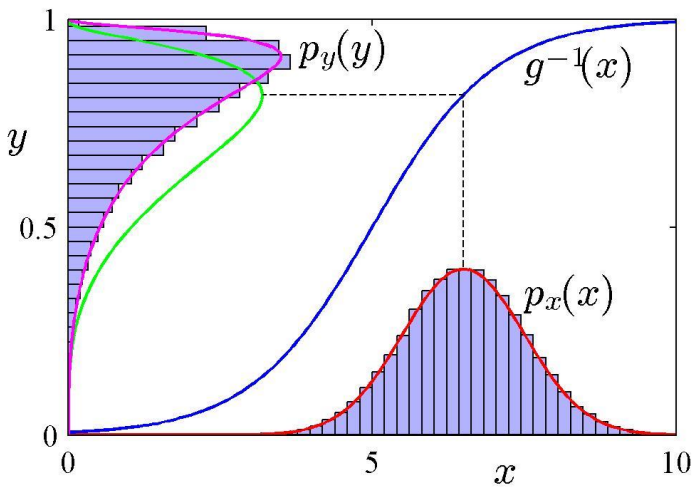
- *Continuous random variables*
- Change of variables

$$x = g(y) \quad dx = g'(y)dy$$

$$p_y(y)dy = p_x(x)dx$$

$$= p_x(g(y))g'(y)dy$$

$$p_y(y) = p_x(g(y))|g'(y)|$$



- Continuous random variables
- Change of variables

$x = g(y) \quad dx = g'(y)dy$

$p_y(y)dy = p_x(x)dx$

$= p_x(g(y))g'(y)dy$

$p_y(y) = p_x(g(y))|g'(y)|$

Probability Theory Review

- *Expectation* of $f(x)$ is an average value of f under some probability distribution $p(x)$

$$\mathbb{E}[f] = \sum_x p(x)f(x) \qquad \mathbb{E}[f] = \int p(x)f(x)dx$$

$$\mathbb{E}[f] \simeq \frac{1}{N}\sum_{n=1}^{N} f(x_n)$$

- *Conditional expectation* wrt. conditional distribution

$$\mathbb{E}_x[f|y] = \sum_x p(x|y)f(x)$$

Probability Theory Review

- *Variance* of $f(x)$ is a measure how much variability is there in $f(x)$ around the $\mathbb{E}[f(x)]$

$$\mathrm{var}[f] = \mathbb{E}[(f(x) - \mathbb{E}[f(x)])^2] = \mathbb{E}[f(x)^2 - 2(f(x)\mathbb{E}[f(x)]) + \mathbb{E}[f(x)]^2] = \mathbb{E}[f(x)^2] - 2\mathbb{E}[f(x)]^2 + \mathbb{E}[f(x)]^2 = \mathbb{E}[f(x)^2] - \mathbb{E}[f(x)]^2$$

- *Covariance* of x and y expresses the extent to which x and y vary together

$$\mathrm{cov}[x, y] = \mathbb{E}_{x,y}[(x - \mathbb{E}[x])(y - \mathbb{E}[y])]$$
$$= \mathbb{E}_{x,y}[xy] - \mathbb{E}[x]\mathbb{E}[y]$$

Probability Theory Review

- Covariance *matrix* for two vectors of random variables $\boldsymbol{x}$ and $\boldsymbol{y}$

$$\mathrm{cov}[\boldsymbol{x}, \boldsymbol{y}] = \mathbb{E}_{\boldsymbol{x},\boldsymbol{y}}[(\boldsymbol{x} - \mathbb{E}[\boldsymbol{x}])(\boldsymbol{y}^T - \mathbb{E}[\boldsymbol{y}^T])]$$

$$= \mathbb{E}_{\boldsymbol{x},\boldsymbol{y}}[\boldsymbol{x}\boldsymbol{y}^T] - \mathbb{E}[\boldsymbol{x}]\mathbb{E}[\boldsymbol{y}^T]$$

$$\boldsymbol{\Sigma} = \mathrm{cov}[\boldsymbol{x}, \boldsymbol{x}]$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \begin{pmatrix} y_1 & y_2 & y_3 \end{pmatrix} = \begin{pmatrix} x_1 y_1 & x_1 y_2 & x_1 y_3 \\ x_2 y_1 & x_2 y_2 & x_2 y_3 \\ x_3 y_1 & x_3 y_2 & x_3 y_3 \end{pmatrix}$$

Probability Theory Review

- Bayesian probabilities convert prior probability into posterior probability by incorporating the evidence from observed data $\mathcal{D} = \{t_1, \dots, t_N\}$

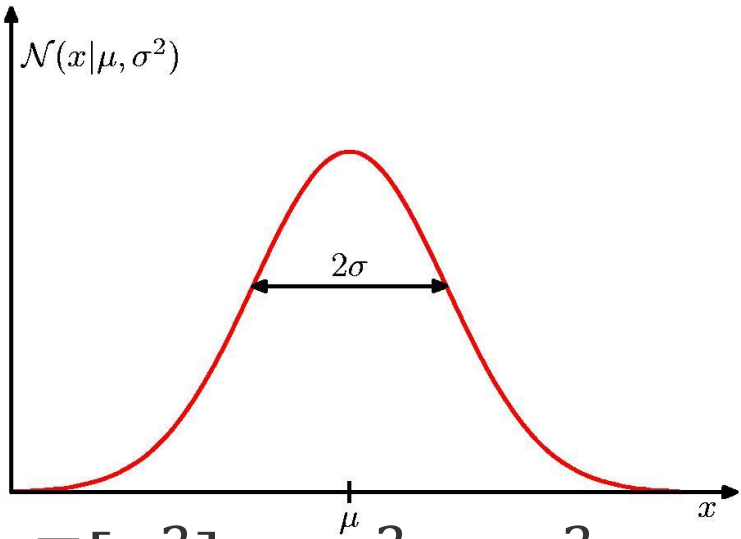
$$p(\boldsymbol{w}|\mathcal{D}) = \frac{p(\mathcal{D}|\boldsymbol{w})p(\boldsymbol{w})}{p(\mathcal{D})} \text{ (posterior} \propto \text{likelihood} \times \text{prior)}$$

- Expresses the uncertainty in $\boldsymbol{w}$ after observing $\mathcal{D}$
- (Maximum) Likelihood function $p(\mathcal{D}|\boldsymbol{w})$

Probability Theory Review

- Gaussian distribution is parameterized by mean μ and variance σ^2

$$\mathcal{N}(x|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



- Standard deviation σ characterizes precision β

$$\beta = \frac{1}{\sigma^2}$$

$$\mathbb{E}[x] = \mu$$

$$\mathbb{E}[x^2] = \mu^2 + \sigma^2$$

$$\mathrm{var}[x] = \mathbb{E}[x]^2 - \mathbb{E}[x^2]$$

Probability Theory Review

- Expectation of Gaussian distribution

$$\mathbb{E}[x] = \int_{-\infty}^{\infty} x \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$z = \frac{x-\mu}{\sigma} \Rightarrow x = \mu + \sigma z, dx = \sigma dz$$

$$\mathbb{E}[x] = \int_{-\infty}^{\infty} (\mu + \sigma z) \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(\mu+\sigma z-\mu)^2}{2\sigma^2}} \sigma dz$$

Probability Theory Review

- Expectation of Gaussian distribution

$$\mathbb{E}[x] = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} (\mu + \sigma z)\, e^{-\frac{z^2}{2}} \sigma dz$$

$$= \frac{1}{\sqrt{2\pi}} \left(\mu \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}} dz + \sigma \int_{-\infty}^{\infty} z e^{-\frac{z^2}{2}} dz \right)$$

$$= \frac{1}{\sqrt{2\pi}} \left(\mu . \sqrt{2\pi} + \sigma . 0 \right) = \mu$$

Probability Theory Review

- Variance of Gaussian distribution

$$\mathbb{E}[x^2] = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} (\mu^2 + 2\mu\sigma z + \sigma^2 z^2) e^{-\frac{z^2}{2}} \sigma dz$$

$$= \frac{1}{\sqrt{2\pi}} \left(\mu^2 \sqrt{2\pi} + 2\mu\sigma . 0 + \sigma^2 \int_{-\infty}^{\infty} z^2 e^{-\frac{z^2}{2}} dz \right)$$

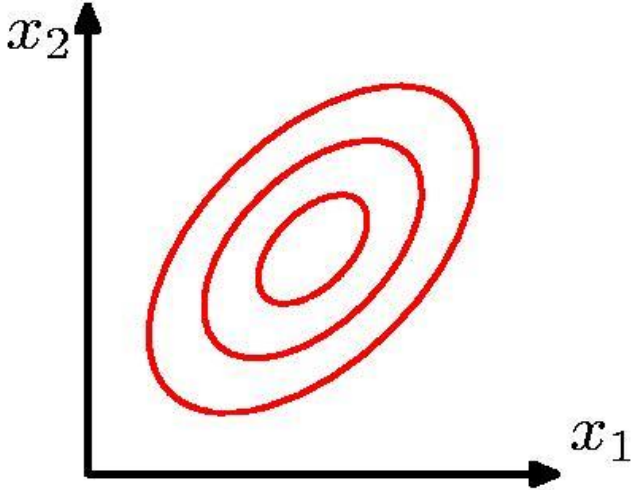
$$= \mu^2 + \frac{1}{\sqrt{2\pi}} \left(\sigma^2 . \sqrt{2\pi} \right) = \mu^2 + \sigma^2$$

Probability Theory Review

- D-dimensional Gaussian distribution

$$\mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu},\boldsymbol{\Sigma}) = \frac{1}{2\pi^{\frac{D}{2}}}\frac{1}{|\boldsymbol{\Sigma}|^{\frac{1}{2}}}\exp\left\{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right\}$$

- $|\boldsymbol{\Sigma}|$ is the determinant of covariance matrix



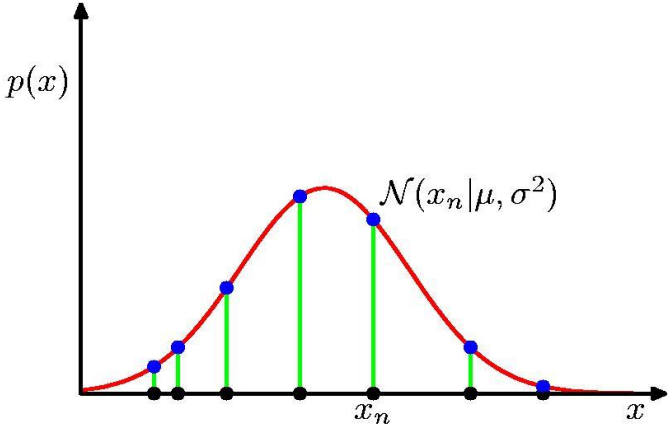
Probability Theory Review

- Likelihood function for Gaussian distribution

$$p(\mathbf{x}|\mu,\sigma^2) = \prod_{n=1}^{N} \mathcal{N}(x|\mu,\sigma^2)$$

- Independent and identically distributed set of observations

$$\mathbf{x} = (x_1 \cdots x_N)^{\mathrm{T}}$$



Probability Theory Review

- Log likelihood

$$\ln p(\mathbf{x}|\mu,\sigma^2) = \ln \prod_{n=1}^{N} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_n-\mu)^2}{2\sigma^2}} =$$

$$\ln \frac{1}{2\pi\sigma^{2\frac{N}{2}}} e^{-\sum_{n=1}^{N}\frac{(x_n-\mu)^2}{2\sigma^2}}$$

$$= -\frac{N}{2}\ln\sigma^2 - \frac{N}{2}\ln(2\pi) - \frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2$$

Probability Theory Review

- Maximum Likelihood Estimators for μ and σ^2
- Find extrema of log likelihood by searching where its partial derivative =0

$$\frac{\partial}{\partial \mu}\left(-\frac{N}{2}\ln\sigma^2 - \frac{N}{2}\ln(2\pi) - \frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2\right) = 0$$

$$-\frac{1}{2\sigma^2}\sum_{n=1}^{N} -2(x_n-\mu) \ = 0$$

$$\frac{1}{\sigma^2}\sum_{n=1}^{N}(x_n-\mu) \ = 0 \Longrightarrow \sum_{n=1}^{N} x_n - \mu N = 0 \Longrightarrow \mu_{ML} = \frac{1}{N}\sum_{n=1}^{N} x_n$$

Probability Theory Review

- Maximum Likelihood Estimators for μ and σ^2
- Find extrema of log likelihood by searching where its partial derivative equals zero

$$\frac{\partial}{\partial \sigma^2}\left(-\frac{N}{2}\ln\sigma^2 - \frac{N}{2}\ln(2\pi) - \frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2\right) = 0$$

$$-\frac{N}{2\sigma^2} - \frac{1}{2}\sum_{n=1}^{N}(x_n-\mu)^2\frac{\partial}{\partial \sigma^2}\left(\frac{1}{\sigma^2}\right) = -\frac{N}{2\sigma^2} - \frac{1}{2}\sum_{n=1}^{N}(x_n-\mu)^2\left(-\frac{1}{\sigma^4}\right) = 0$$

$$\frac{1}{2\sigma^2}\left(\frac{1}{\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2 - N\right) = 0 \Longrightarrow \sum_{n=1}^{N}(x_n-\mu)^2 = \sigma^2 N \Longrightarrow \sigma^2{}_{ML} = \frac{1}{N}\sum_{n=1}^{N}(x_n-\mu_{ML})^2$$

Probability Theory Review

- Expectation for μ_{ML} and $\sigma^2{}_{ML}$

$$\mathbb{E}[\mu_{ML}] = \mathbb{E}\left[\frac{1}{N}\sum_{n=1}^{N} x_n\right] = \frac{1}{N}\sum_{n=1}^{N} \mathbb{E}[x_n] =$$

$$\frac{1}{N}\sum_{n=1}^{N} \mu = \mu$$

Probability Theory Review

- Expectation for μ_{ML} and $\sigma^2{}_{ML}$

$$\mathbb{E}[\sigma^2{}_{ML}] = \mathbb{E}\left[\frac{1}{N}\sum_{n=1}^{N}(x_n - \mu + \mu - \mu_{ML})^2\right]$$

$$(x_n - \mu + \mu - \mu_{ML})^2$$
$$= (x_n - \mu)^2 + 2(x_n - \mu)(\mu - \mu_{ML}) + (\mu - \mu_{ML})^2$$

Probability Theory Review

- Expectation for μ_{ML} and $\sigma^2{}_{ML}$

$$\sum_{n=1}^{N}(x_n-\mu_{ML})^2$$

$$=\sum_{n=1}^{N}(x_n-\mu)^2+2(\mu-\mu_{ML})\sum_{n=1}^{N}(x_n-\mu)+\sum_{n=1}^{N}(\mu-\mu_{ML})^2$$

$$\sum_{n=1}^{N}x_n-N\mu=N\frac{1}{N}\sum_{n=1}^{N}x_n-N\mu=N\mu_{ML}-N\mu=N(\mu_{ML}-\mu)$$

Probability Theory Review

- Expectation for μ_{ML} and $\sigma^2{}_{ML}$

$$\sum_{n=1}^{N}(x_n - \mu_{ML})^2$$

$$= \sum_{n=1}^{N}(x_n - \mu)^2 - 2N(\mu - \mu_{ML})^2 + N(\mu - \mu_{ML})^2$$

$$= \sum_{n=1}^{N}(x_n - \mu)^2 - N(\mu - \mu_{ML})^2$$

Probability Theory Review

- Expectation for μ_{ML} and $\sigma^2{}_{ML}$

$$\mathbb{E}[\sigma^2{}_{ML}] = \mathbb{E}\left[\frac{1}{N}\sum_{n=1}^{N}(x_n - \mu)^2\right] - \mathbb{E}[(\mu_{ML} - \mu)^2]$$

$$= \frac{1}{N}\sum_{n=1}^{N}\mathbb{E}[(x_n - \mu)^2] - \mathrm{Var}(\mu_{ML}) = \frac{1}{N}\sum_{n=1}^{N}\sigma^2 - \mathrm{Var}\left(\frac{1}{N}\sum_{n=1}^{N}x_n\right)$$

Probability Theory Review

- Expectation for μ_{ML} and $\sigma^2{}_{ML}$

$$\mathrm{Var}(cX) = c^2\mathrm{Var}(X)$$

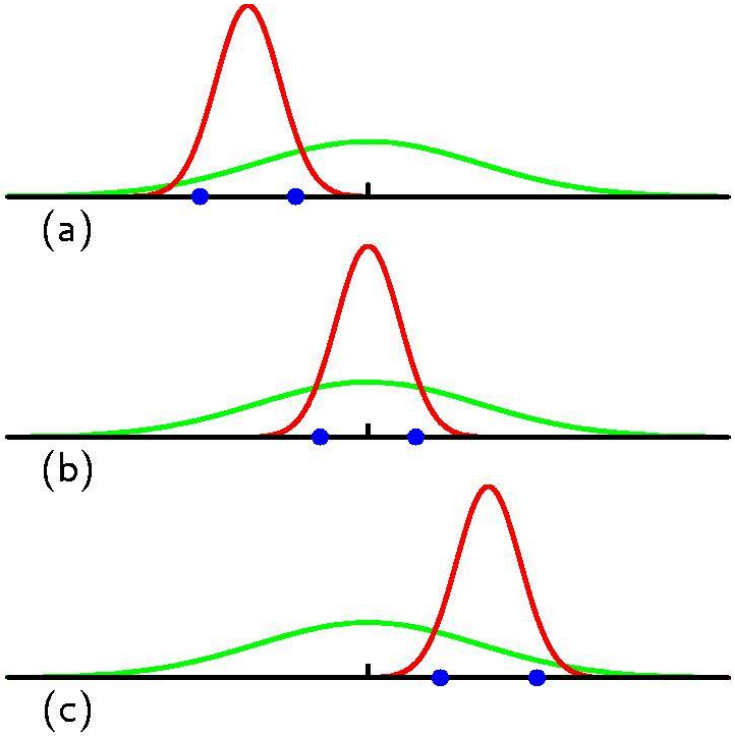
$$\mathrm{Var}(cX) = \mathbb{E}[(cX - \mathbb{E}[cX])^2] = \mathbb{E}[(cX - c\mathbb{E}[X])^2]$$
$$= \mathbb{E}[c^2(X - \mathbb{E}[X])^2] = c^2\mathbb{E}[(X - \mathbb{E}[X])^2]$$
$$= c^2\mathrm{Var}(X)$$

Probability Theory Review

- Expectation for μ_{ML} and $\sigma^2{}_{ML}$

$$\mathbb{E}[\sigma^2{}_{ML}] = \sigma^2 - \frac{1}{N^2}\sum_{n=1}^{N}\mathrm{Var}(x_n)$$

$$= \sigma^2 - \frac{N\sigma^2}{N^2} = \frac{N-1}{N}\sigma^2$$



Probability Theory Review

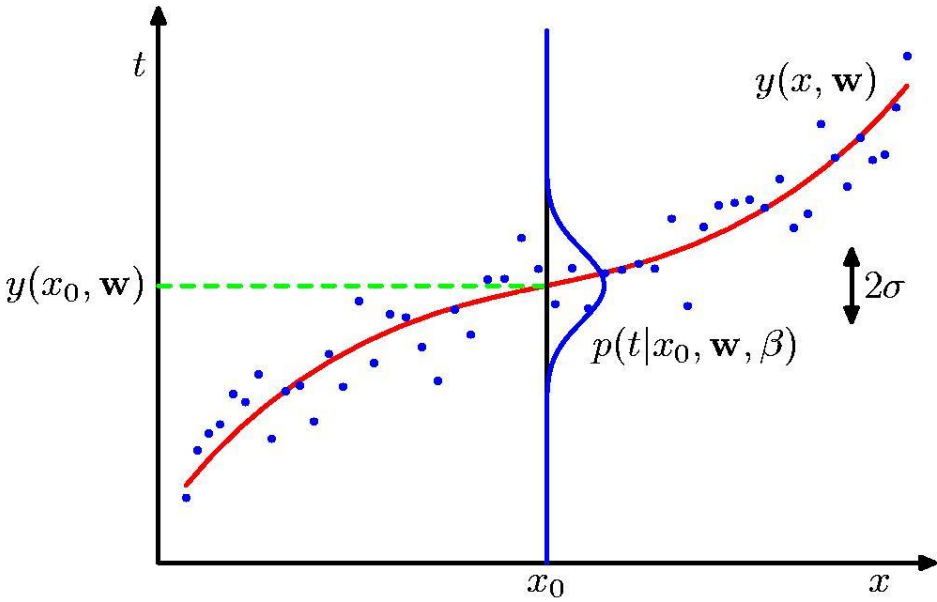
- Unbiased estimate of variance $\hat{\sigma}^2$

$$\hat{\sigma}^2 = \frac{N}{N-1}\sigma^2{}_{ML} = \frac{1}{N-1}\sum_{n=1}^{N}(x_n - \mu_{ML})^2$$

Polynomial Curve Fitting

- Probabilistic perspective assumes Gaussian distribution with $p(t|x, \boldsymbol{w}, \beta) = \mathcal{N}(t|y(x, \boldsymbol{w}), \beta^{-1})$

$$p(\mathbf{t}|\mathbf{x}, \boldsymbol{w}, \beta) = \prod_{n=1}^{N} \mathcal{N}(t_n|y(x_n, \boldsymbol{w}), \beta^{-1})$$



Polynomial Curve Fitting

- Minimizing negative log likelihood with respect to $\boldsymbol{w}$ (and replacing multiplier β) leads to sum-of-squares error function:

$$-\ln p(\mathbf{t}|\mathbf{x}, \boldsymbol{w}, \beta) = -\frac{N}{2}\ln\beta + \frac{N}{2}\ln(2\pi) + \frac{\beta}{2}\sum_{n=1}^{N}\{y(x_n, \boldsymbol{w}) - t_n\}^2$$

$$\boldsymbol{w}_{ML} = \arg\min_{\boldsymbol{w}}\left(\frac{1}{2}\sum_{n=1}^{N}\{y(x_n, \boldsymbol{w}) - t_n\}^2\right)$$

Polynomial Curve Fitting

- Maximizing log likelihood with respect to β:

$$\ln p(\mathbf{t}|\mathbf{x}, \boldsymbol{w}, \beta) = \frac{N}{2}\ln\beta - \frac{N}{2}\ln(2\pi) - \frac{\beta}{2}\sum_{n=1}^{N}\{y(x_n, \boldsymbol{w}) - t_n\}^2$$

$$\frac{\partial}{\partial\beta}(\ln p(\mathbf{t}|\mathbf{x}, \boldsymbol{w}, \beta)) = \frac{N}{2\beta} - \frac{1}{2}\sum_{n=1}^{N}\{y(x_n, \boldsymbol{w}) - t_n\}^2 = 0$$

$$\frac{1}{\beta_{ML}} = \frac{1}{N}\sum_{n=1}^{N}\{y(x_n, \boldsymbol{w}_{ML}) - t_n\}^2$$

- Predictive distribution:

$$p(t|x, \boldsymbol{w}_{ML}, \beta_{ML}) = \mathcal{N}\left(t\middle|y(x, \boldsymbol{w}_{ML}), {\beta_{ML}}^{-1}\right)$$

Polynomial Curve Fitting

- Prior distribution over polynomial coefficients $\boldsymbol{w}$:

$$p(\boldsymbol{w}|\alpha) = \mathcal{N}(\boldsymbol{w}|\mathbf{0}, \alpha^{-1}\mathbf{I})$$

$$= \frac{1}{2\pi^{\frac{M+1}{2}}} \frac{1}{|\alpha^{-1}\mathbf{I}|^{\frac{1}{2}}} \exp\left\{-\frac{1}{2}(\boldsymbol{w}-\mathbf{0})^T \alpha \mathbf{I}(\boldsymbol{w}-\mathbf{0})\right\} =$$

$$\left(\frac{\alpha}{2\pi}\right)^{\frac{M+1}{2}} \exp\left\{-\frac{\alpha}{2}\boldsymbol{w}^T\boldsymbol{w}\right\}$$

- Posterior distribution for $\boldsymbol{w}$:

$$p(\boldsymbol{w}|\mathbf{x}, \mathbf{t}, \beta) \propto p(\mathbf{t}|\mathbf{x}, \boldsymbol{w}, \beta).\, p(\boldsymbol{w}|\alpha)$$

Polynomial Curve Fitting

- Maximum Posterior (MAP) for $\boldsymbol{w}$:

$$-\ln\big(p(\mathbf{t}|\mathbf{x},\boldsymbol{w},\beta).p(\boldsymbol{w}|\alpha)\big) = -\ln\big(p(\mathbf{t}|\mathbf{x},\boldsymbol{w},\beta)\big) - \ln\big(p(\boldsymbol{w}|\alpha)\big)$$

$$= -\frac{N}{2}\ln\beta + \frac{N}{2}\ln(2\pi) + \frac{\beta}{2}\sum_{n=1}^{N}\{y(x_n,\boldsymbol{w}) - t_n\}^2 - \frac{M+1}{2}\ln\frac{\alpha}{2\pi} + \frac{\alpha}{2}\boldsymbol{w}^T\boldsymbol{w}$$

$$= \frac{\beta}{2}\sum_{n=1}^{N}\{y(x_n,\boldsymbol{w}) - t_n\}^2 + \frac{\alpha}{2}\boldsymbol{w}^T\boldsymbol{w} + const.$$

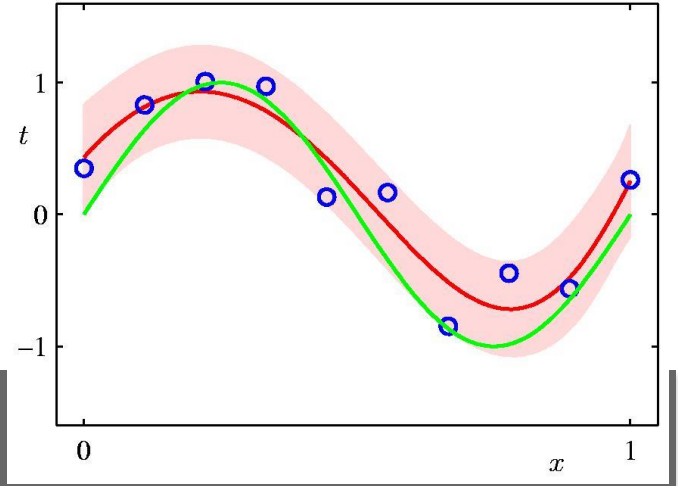
$$\boldsymbol{w}_{MAP} = \arg\min_{\boldsymbol{w}}\left(\frac{\beta}{2}\sum_{n=1}^{N}\{y(x_n,\boldsymbol{w}) - t_n\}^2 + \frac{\alpha}{2}\boldsymbol{w}^T\boldsymbol{w}\right)$$

Polynomial Curve Fitting

- Bayesian curve fitting: instead of point estimate we evaluate the predictive distribution:

$$p(t|x, \mathbf{x}, \mathbf{t}, \alpha, \beta) = \int p(t|x, \boldsymbol{w}, \beta).\, p(\boldsymbol{w}|\mathbf{x}, \mathbf{t}, \alpha, \beta) d\boldsymbol{w}$$

$$= \mathcal{N}(t|m(x), s^2(x))$$

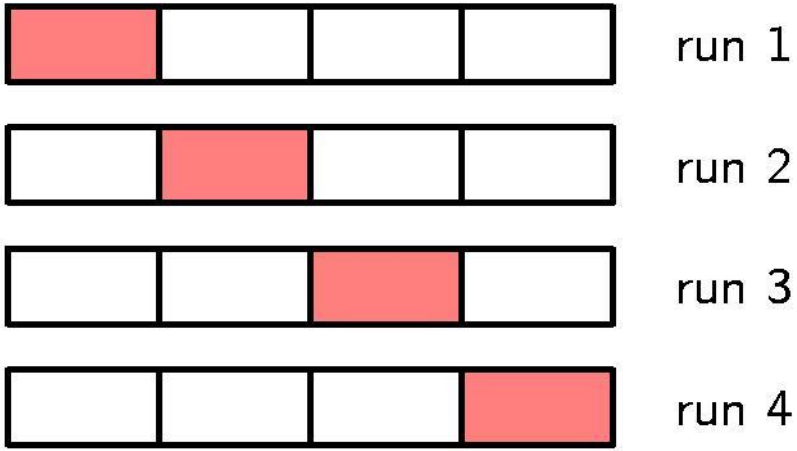


Model Selection

- Polynomial curve fitting has a M determines the polynomial degree
- Regularized least squares have λ controls the complexity
- Parameter choice governed by the best predictive performance
- When plenty of data is available, train many different models on range of values
- Use independent data – *validation set* – to select the model with best predictive performance
- To prevent overfitting to validation data, *test set* is reserved for final evaluation

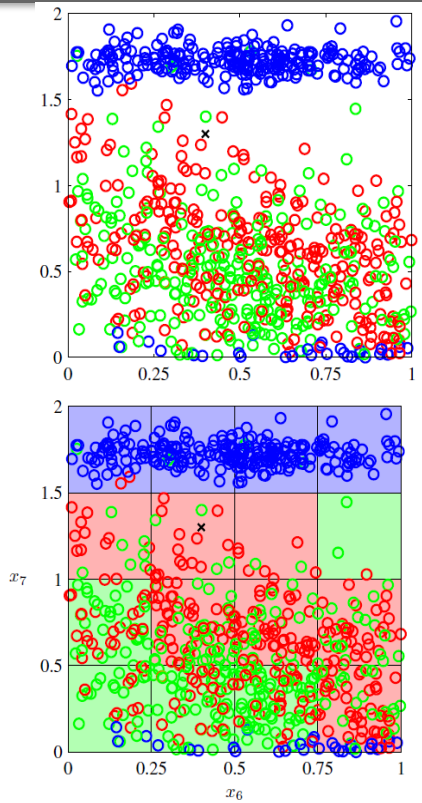
Model Selection

- *Cross-validation* is model selection method for smaller data
- *S-fold* cross-validation uses $(S-1)/S$ data for each model training, performance is assessed on entire data.
- *Leave-one-out* is S-fold cross-validation where $S = N$
- Training duration increase $S \times$
- Number of parameters define S



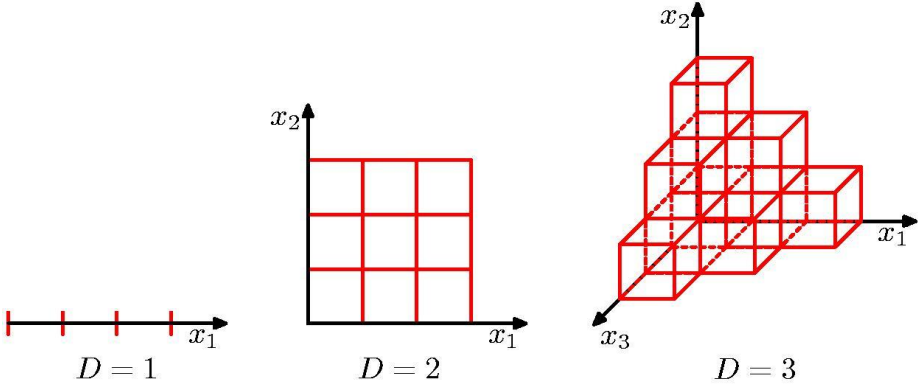
Curse of Dimensionality

- Example of oil flow data with 12 variables
- RGB Color codes ‘homogeneous’, ‘annular’, ‘laminar’ categories
- Predicting category for point x based on nearby data points
- Uniform space subdivision for category cells



Curse of Dimensionality

- With increasing dimensionality space becomes more sparse
- To keep the same density, data would need to grow exponentially



Curse of Dimensionality

- In polynomial curve fitting increasing dimensionality leads to more coefficients w
- Polynomial of order 3:

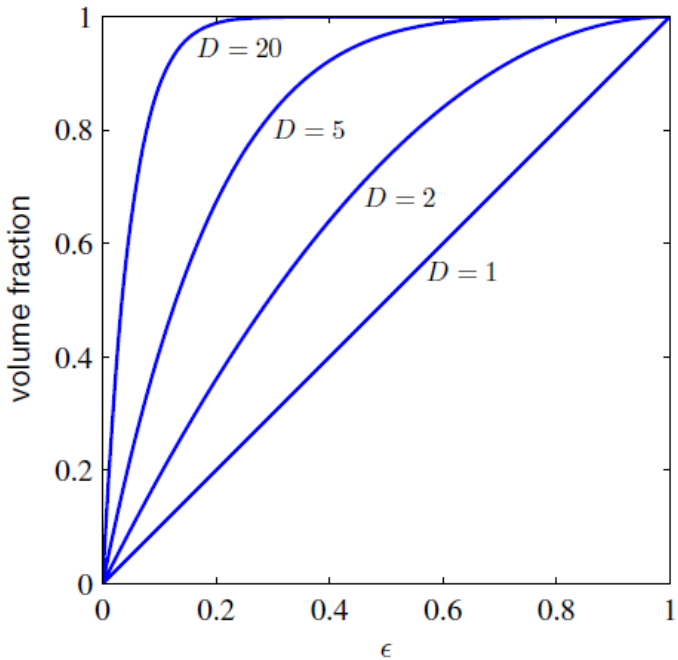
$$y(\boldsymbol{x}, \boldsymbol{w}) = w_0 + \sum_{i=1}^{D} w_i x_i + \sum_{i=1}^{D}\sum_{j=1}^{D} w_{ij} x_i x_j + \sum_{i=1}^{D}\sum_{j=1}^{D}\sum_{k=1}^{D} w_{ijk} x_i x_j x_k$$

- Power law growth for polynomial of order M dimensionality determines number coefficients as D^M

Curse of Dimensionality

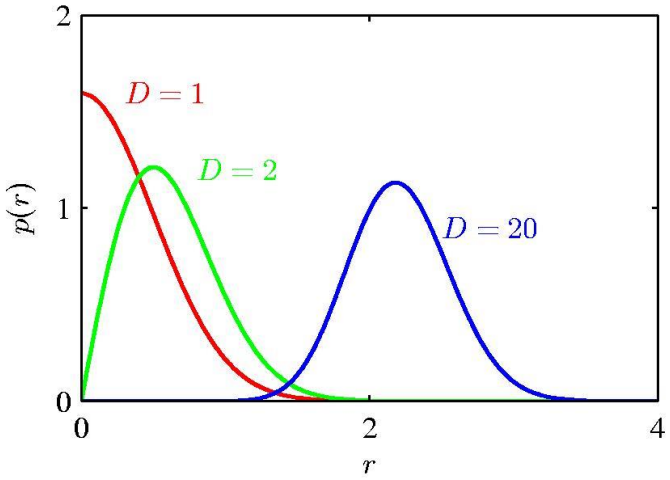
- Unit sphere of various dimensions D
- Fraction of volume of the sphere between $r \in [1-\epsilon, 1]$

$$\frac{V_D(1) - V_D(1-\epsilon)}{V_D(1)} = 1 - (1-\epsilon)^D$$



Curse of Dimensionality

- Gaussian distribution in higher dimensions D
- Probability mass distribution shifts with increasing dimensions away from 0



Curse of Dimensionality

- Even in high-dimensional spaces machine learning techniques provide effective solutions
- The data has often lower effective dimensionality and directions over which it changes
- Data is often smooth with small changes in the target variables

Curse of Dimensionality

- Example: Automatic planar object detection
- Conveyor belt imaging results in many pixels (512x512 dimensions), but the outcome is only position and rotation – three degrees of freedom of variability
- Set of images will live on a three-dimensional non-linear manifold embedded within the high-dimensional space

Decision Theory

- *Inference* is determination of joint probability distribution of $p(\mathbf{x}, \mathbf{t})$ from input vector $\mathbf{x}$ together with a target vector $\mathbf{t}$
- The joint probability distribution $p(\mathbf{x}, \mathbf{t})$ characterizes uncertainty associated with these variables
- *Decision theory* determines optimal value of $\mathbf{t}$ or takes a specific action based on our understanding of the values $\mathbf{t}$

Decision Theory

- Cancer diagnosis from X-ray:
- Input vector $\boldsymbol{x}$ of pixels and output t is presence $\mathcal{C}_1$ $(t = 0)$ or absence $\mathcal{C}_2$ of cancer $(t = 1)$
- Inference determines joint distribution $p(\boldsymbol{x}, \mathcal{C}_k)$
- Decision step defines the treatment plan

Decision Theory

- For a new image $\boldsymbol{x}$
- Bayes theorem expresses the probabilities of the two classes given an image:

$$p(\mathcal{C}_k|\boldsymbol{x}) = \frac{p(\boldsymbol{x}|\mathcal{C}_k)p(\mathcal{C}_k)}{p(\boldsymbol{x})}$$

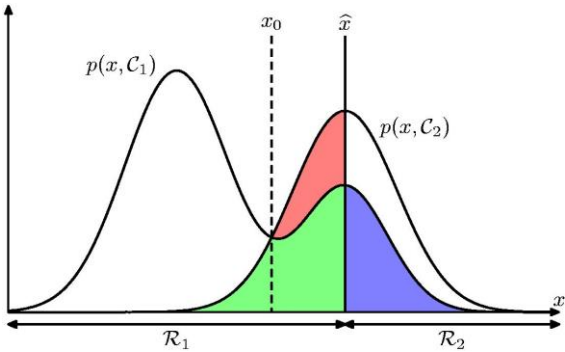
- Prior probability of person having cancer $p(\mathcal{C}_k)$ and $p(\mathcal{C}_k|\boldsymbol{x})$ is the posterior probability

Decision Theory

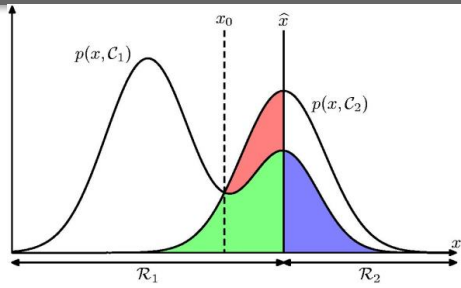
- Minimizing misclassification rate
- *Decision regions* $\mathcal{R}_k$ and decision boundaries

$$p(\text{mistake}) = p(\boldsymbol{x} \in \mathcal{R}_1, \mathcal{C}_2) + p(\boldsymbol{x} \in \mathcal{R}_2, \mathcal{C}_1)$$

$$= \int_{\mathcal{R}_1} p(\boldsymbol{x}, \mathcal{C}_2) d\boldsymbol{x} + \int_{\mathcal{R}_2} p(\boldsymbol{x}, \mathcal{C}_1) d\boldsymbol{x}$$



Decision Theory



- Maximizing probability of being correct
- $p(\text{correct}) =$

$$\sum_{k=1}^{K} p(\boldsymbol{x} \in \mathcal{R}_k, \mathcal{C}_k) = \sum_{k=1}^{K} \int_{\mathcal{R}_k} p(\boldsymbol{x}, \mathcal{C}_k) d\boldsymbol{x}$$

- As $p(\boldsymbol{x})$ is in all terms minimum probability of mistake / maximum probability of correctness is obtained if each $\boldsymbol{x}$ is assigned to the class with higher $p(\mathcal{C}_k|\boldsymbol{x})$

Decision Theory

- Minimizing expected loss
- Misclassifying healthy as having cancer is much less fatal as classifying patient with cancer as healthy
- It is better to make more mistakes of first kind than of second kind
- Formalizing done through *cost / loss functions* as overall measure of loss taking any available actions
- Optimal solution is to minimize the total loss

Decision Theory

- Minimizing expected loss
- L_{kj} expresses the loss when $\boldsymbol{x}$ belonging to $\mathcal{C}_k$ is assigned to $\mathcal{C}_j$
- In *loss matrix* is k,j element
- However true class is unknown and its uncertainty is quantified through joint probability distribution

$$\begin{array}{cc} & \text{Decision} \\ & \begin{array}{cc} \text{cancer} & \text{normal} \end{array} \\ \text{Truth} \begin{array}{c} \text{cancer} \\ \text{normal} \end{array} & \begin{pmatrix} 0 & 1000 \\ 1 & 0 \end{pmatrix} \end{array}$$

Decision Theory

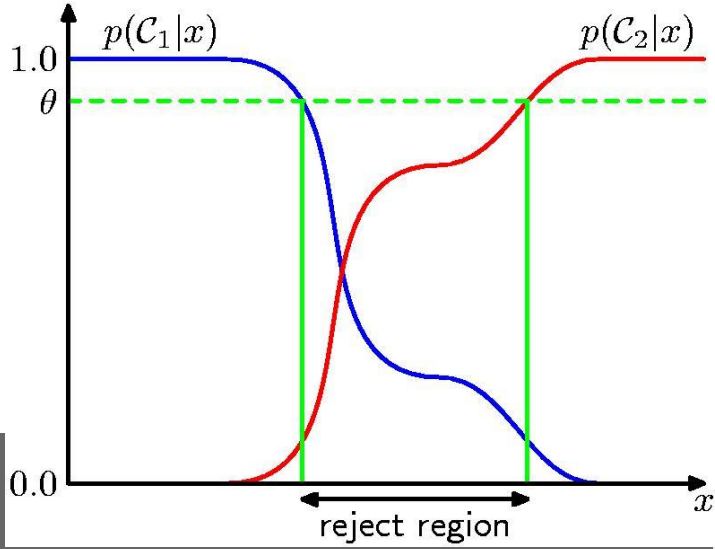
- Minimizing expected loss
- Therefore we minimize average loss by choosing $\mathcal{R}_j$ regions:

$$\mathbb{E}[L] = \sum_k \sum_j \int_{\mathcal{R}_j} L_{kj} p(\boldsymbol{x}, \mathcal{C}_k) d\boldsymbol{x}$$

This corresponds to assigning each $\boldsymbol{x}$ to class j for which $\sum_k L_{kj} p(\mathcal{C}_k | \boldsymbol{x})$ is a minimum.

Decision Theory

- The most misclassifications come from regions where $p(\mathcal{C}_k|\boldsymbol{x})$ significantly below unity
- *Reject option* avoids making decisions on difficult cases



Decision Theory

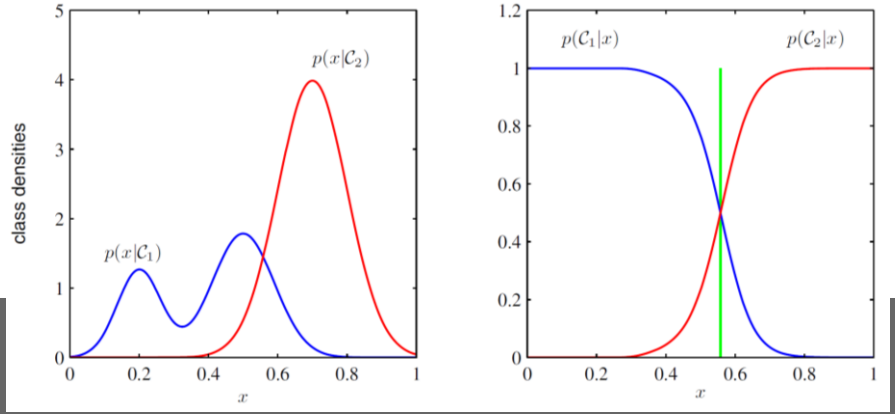
- *Generative models: inference stage* learns $p(\mathcal{C}_k|\boldsymbol{x})$

$$p(\mathcal{C}_k|\boldsymbol{x}) = \frac{p(\boldsymbol{x}|\mathcal{C}_k)p(\mathcal{C}_k)}{\sum_{k=1}^{K} p(\boldsymbol{x}|\mathcal{C}_k)p(\mathcal{C}_k)}$$

- *Decision stage* models explicitly or implicitly joint distribution of inputs and outputs $p(\boldsymbol{x}, \mathcal{C}_k)$
- Allows for generating synthetic data in input space
- Most demanding type of models

Decision Theory

- *Discriminative models*: learn $p(\mathcal{C}_k|\boldsymbol{x})$ and use decision theory to assign $\boldsymbol{x}$ to one of the classes
- Directly find a *discriminant function* $f(\boldsymbol{x})$ that maps $\boldsymbol{x}$ onto a label, probabilities play no role



Decision Theory

- For risk minimization if loss matrix changes, discriminant function would need to be recomputed from scratch
- Posterior probability $p(\mathcal{C}_k|\boldsymbol{x})$ can be updated with new data
- Reject option can be used if $p(\mathcal{C}_k|\boldsymbol{x})$ is available
- Compensating for class priors if we have unbalanced dataset of many healthy and few pathological cases

Decision Theory

- Combining models: X-ray $\boldsymbol{x}_I$, blood samples $\boldsymbol{x}_B$ are independent random variables
- *Conditional independence (naive Bayes model)*

$$p(\boldsymbol{x}_I, \boldsymbol{x}_B | \mathcal{C}_k) = p(\boldsymbol{x}_I | \mathcal{C}_k).p(\boldsymbol{x}_B | \mathcal{C}_k)$$

$$p(\mathcal{C}_k | \boldsymbol{x}_I, \boldsymbol{x}_B) \propto p(\boldsymbol{x}_I, \boldsymbol{x}_B | \mathcal{C}_k) p(\mathcal{C}_k)$$

$$p(\mathcal{C}_k | \boldsymbol{x}_I, \boldsymbol{x}_B) \propto p(\boldsymbol{x}_I | \mathcal{C}_k).p(\boldsymbol{x}_B | \mathcal{C}_k) p(\mathcal{C}_k)$$

$$p(\mathcal{C}_k | \boldsymbol{x}_I, \boldsymbol{x}_B) \propto \frac{p(\mathcal{C}_k | \boldsymbol{x}_I).p(\mathcal{C}_k | \boldsymbol{x}_B)}{p(\mathcal{C}_k)}$$

Decision Theory

- Loss functions for regression

$$\mathbb{E}[L] = \int_x \int_t L(t, y(\boldsymbol{x}))p(\boldsymbol{x}, t)d\boldsymbol{t}d\boldsymbol{x}$$

$$\mathbb{E}[L] = \int_x \int_t \{y(\boldsymbol{x}) - t\}^2 p(\boldsymbol{x}, t)d\boldsymbol{t}d\boldsymbol{x}$$

Decision Theory

- Loss functions for regression

$$\frac{\delta \mathbb{E}[L]}{\delta \boldsymbol{y}(\boldsymbol{x})} = 2\int_t \{y(\boldsymbol{x}) - t\}p(\boldsymbol{x}, t)dt = 0$$

$$\int_t y(\boldsymbol{x})p(\boldsymbol{x}, t)dt = \int_t tp(\boldsymbol{x}, t)dt$$

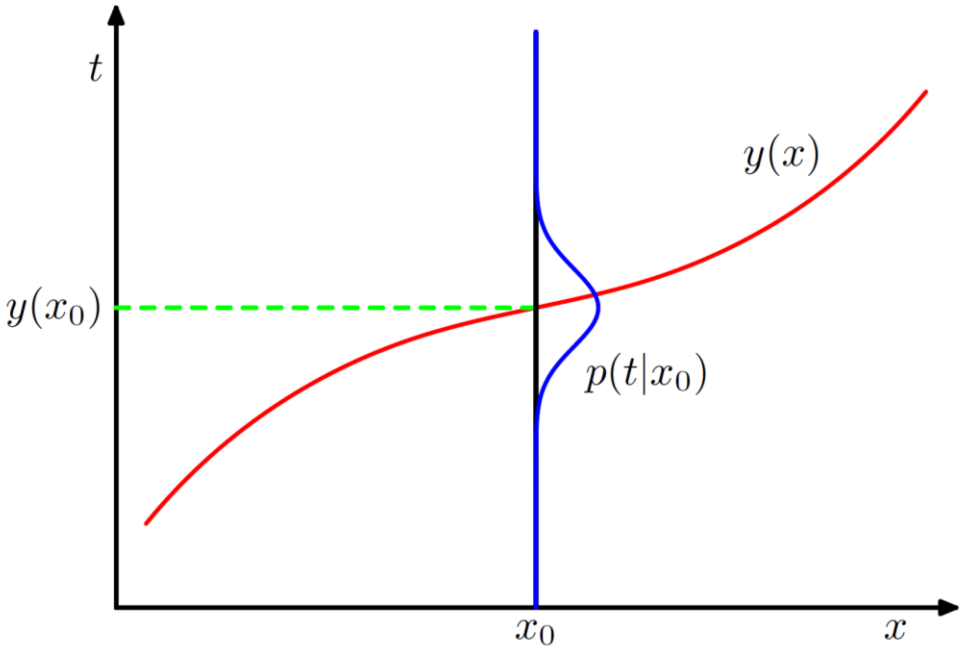
$$y(\boldsymbol{x})p(\boldsymbol{x}) = \int_t tp(\boldsymbol{x}, t)dt$$

Decision Theory

- Loss functions for regression

$$y(\boldsymbol{x}) = \frac{\int_t tp(\boldsymbol{x}, t)dt}{p(\boldsymbol{x})} =$$

$$\int_t tp(t|\boldsymbol{x})dt = \mathbb{E}_t[t|\boldsymbol{x}]$$



Information Theory

- Discrete random variable x
- Information value: the degree of surprise on learning the value of x
- Occurrence of highly improbable event leads to new information
- Highly probable event does not lead to new information

Information Theory

- Information $h(x)$ depends on probability $p(x)$
- For x, y independent events happen the information content $h(x, y) = h(x) + h(y)$
- The joint probability of independent random variables x, y: $p(x, y) = p(x)p(y)$
- Information content $h(x) = -\log_2(p(x))$

Information Theory

- Entropy $H[x]$ is quantity measuring average amount of information as an expectation of $h(x)$ wrt. $p(x)$

$$H[x] = -\sum_{x} p(x)\log_2(p(x))$$

Information Theory

- Example: x is a discrete random variable with 8 uniformly distributed states

$$H[x] = -8\left(\frac{1}{8}\log_2\frac{1}{8}\right) = 3\ bits$$

Information Theory

- Example: x is a discrete random variable with states (a, b, c, d, e, f, g, h) occurring with probability $\left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{64}, \frac{1}{64}, \frac{1}{64}, \frac{1}{64}\right)$

$$H[x]$$

$$= -\left(\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{4}\log_2\frac{1}{4} + \frac{1}{8}\log_2\frac{1}{8} + \frac{1}{16}\log_2\frac{1}{16} + \frac{1}{64}\log_2\frac{1}{64}\right.$$

$$\left. + \frac{1}{64}\log_2\frac{1}{64} + \frac{1}{64}\log_2\frac{1}{64} + \frac{1}{64}\log_2\frac{1}{64}\right) = 2\ bits$$

Information Theory

- Example: x is a discrete random variable with states (a, b, c, d, e, f, g, h) occurring with probability $\left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{64}, \frac{1}{64}, \frac{1}{64}, \frac{1}{64}\right)$
- States can be efficiently encoded with strings: $0, 10, 110, 1110, 111100, 111101, 111110, 11111$
- Avg length: $\frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \frac{6}{64} + \frac{6}{64} + \frac{6}{64} + \frac{6}{64} = 2$

Information Theory

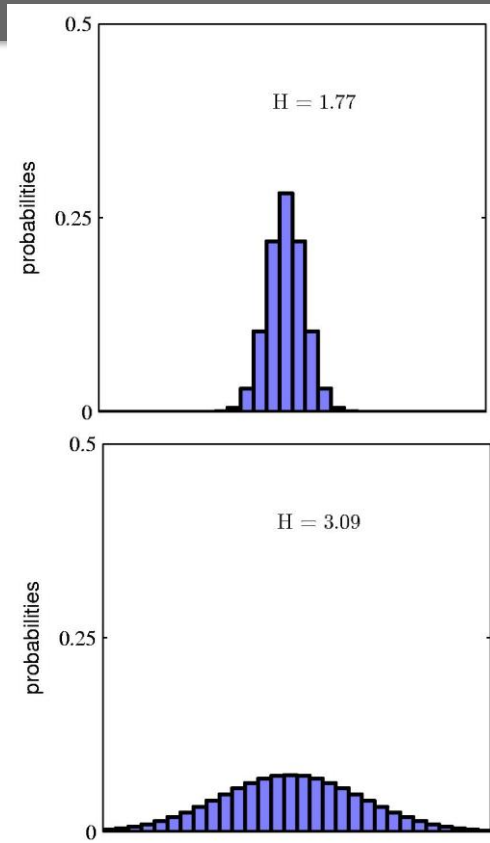
- Entropy in *nats*:

$$H[x] = -\sum_{x} p(x)\ln(p(x))$$

- Maximum entropy by maximizing

$$\widetilde{H}[x] = L(p(x_i), \lambda)$$

$$= -\sum_{i} p(x_i)\ln\big(p(x_i)\big) + \lambda\left(\sum_{x} p(x_i) - 1\right)$$



Information Theory

$$\frac{\partial L}{\partial p(x_i)} = -\left(\ln\left(p(x_i)\right) + 1\right) + \lambda = 0 \Longrightarrow$$

$$\ln\left(p(x_i)\right) = \lambda - 1 \Longrightarrow p(x_i) = e^{\lambda - 1}$$

$$\sum_{i=1}^{N} p(x_i) = 1 = \sum_{i=1}^{N} e^{\lambda - 1} = N e^{\lambda - 1}$$

$$e^{\lambda - 1} = \frac{1}{N} = p(x_i)$$

Information Theory

- *Differential entropy*: extension of entropy for continuous variable x
- Dividing into bins of width Δ and using *mean-value theorem*

$$\int_{i\Delta}^{(i+1)\Delta} p(x)dx = p(x_i)\Delta$$

- Continuous variable x is assigned to x_i when it belongs to ith bin with $p(x_i)\Delta$

Information Theory

- Differential entropy:

$$H_\Delta = -\sum_i p(x_i)\Delta \ln(p(x_i)\Delta)$$

$$= -\sum_i p(x_i)\Delta \ln\big(p(x_i)\big) - \ln(\Delta)$$

$$\lim_{\Delta\to 0}\left\{-\sum_i p(x_i)\Delta \ln\big(p(x_i)\big)\right\} = -\int p(x)\ln\big(p(x)\big) = H[x]$$

Information Theory

- Maximizing differential entropy with constraints:

$$\int_{-\infty}^{\infty} p(x)dx = 1$$

$$\int_{-\infty}^{\infty} xp(x)dx = \mu$$

$$\int_{-\infty}^{\infty} (x-\mu)^2 p(x)dx = \sigma^2$$

Information Theory

- Maximizing differential entropy with constraints:

$$\frac{\partial}{\partial p(x)}\left[-\int p(x)\ln\big(p(x)\big)\,dx+\lambda_1\left(\int_{-\infty}^{\infty}p(x)dx-1\right)\right.$$

$$\left.+\lambda_2\left(\int_{-\infty}^{\infty}xp(x)dx-\mu\right)+\lambda_3\left(\int_{-\infty}^{\infty}(x-\mu)^2p(x)dx-\sigma^2\right)\right]$$

$$=0$$

Information Theory

- Maximizing differential entropy with constraints:

$$\ln\big(p(x)\big) + 1 - \lambda_1 - \lambda_2 x - \lambda_3 (x-\mu)^2 = 0$$

$$\Rightarrow p(x) = e^{-1+\lambda_1+\lambda_2 x+\lambda_3 (x-\mu)^2}$$

$$= A \exp(\lambda_2 x) \exp(\lambda_3 (x-\mu)^2)$$

- To satisfy finite mean and variance $\lambda_2 = 0$

- $\int_{-\infty}^{\infty} \exp(\lambda_3 (x-\mu)^2)\, dx = \sqrt{\frac{\pi}{-\lambda_3}} \Rightarrow A = \sqrt{\frac{-\lambda_3}{\pi}}$

Information Theory

$$\int_{-\infty}^{\infty} (x-\mu)^2 p(x) dx$$

$$= \int_{-\infty}^{\infty} (x-\mu)^2 \sqrt{\frac{-\lambda_3}{\pi}} \exp(\lambda_3 (x-\mu)^2)\, dx = \frac{1}{2(-\lambda_3)} = \sigma^2$$

- Therefore substituting for Lagrange multipliers leads to:

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Information Theory

- Maximizing differential entropy:

$$H[x] = -\int p(x)\ln\big(p(x)\big)dx$$

$$= -\int p(x)\left\{\ln\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \frac{(x-\mu)^2}{2\sigma^2}\right\}dx$$

$$= -\int p(x)\left\{-\frac{1}{2}\ln(2\pi\sigma^2)\right\}dx - \int p(x)\left\{\frac{(x-\mu)^2}{2\sigma^2}\right\}dx$$

Information Theory

- Maximizing differential entropy:

$$H[x] = \frac{1}{2}\ln(2\pi\sigma^2)\int p(x)dx + \frac{1}{2\sigma^2}\int p(x)(x-\mu)^2 dx$$

$$= \frac{1}{2}\ln(2\pi\sigma^2) + \frac{1}{2\sigma^2}\sigma^2 = \frac{1}{2}(\ln(2\pi\sigma^2) + 1)$$

- With increasing sigma entropy increases
- It can also be negative

Information Theory

- In joint distribution we already know the outcome of $\boldsymbol{x}$
- *Conditional entropy* measures the average additional information needed to specify $\boldsymbol{y}$

$$H[\boldsymbol{y}|\boldsymbol{x}] = \mathbb{E}_{p(\boldsymbol{x},\boldsymbol{y})}[-\ln p(\boldsymbol{y}|\boldsymbol{x})] = -\int\int p(\boldsymbol{x},\boldsymbol{y}) \ln p(\boldsymbol{y}|\boldsymbol{x})\, d\boldsymbol{y} d\boldsymbol{x}$$

$$= \int p(\boldsymbol{x}) \left(-\int p(\boldsymbol{y}|\boldsymbol{x}) \ln p(\boldsymbol{y}|\boldsymbol{x})\, d\boldsymbol{y} \right) d\boldsymbol{x}$$

Information Theory

- Conditional entropy:

$$H[\boldsymbol{x},\boldsymbol{y}] - H[\boldsymbol{y}|\boldsymbol{x}] = -\int\int p(\boldsymbol{x},\boldsymbol{y}) \ln p(\boldsymbol{x},\boldsymbol{y})\, d\boldsymbol{y}d\boldsymbol{x} + \int\int p(\boldsymbol{x},\boldsymbol{y}) \ln p(\boldsymbol{y}|\boldsymbol{x})\, d\boldsymbol{y}d\boldsymbol{x}$$

$$= -\int\int p(\boldsymbol{x},\boldsymbol{y}) \ln\left(\frac{p(\boldsymbol{x},\boldsymbol{y})}{p(\boldsymbol{y}|\boldsymbol{x})}\right) d\boldsymbol{y}\, d\boldsymbol{x} = -\int\left(\ln p(\boldsymbol{x}) \int p(\boldsymbol{x},\boldsymbol{y}) d\boldsymbol{y}\right) d\boldsymbol{x}$$

$$= -\int p(\boldsymbol{x}) \ln p(\boldsymbol{x})\, d\boldsymbol{x} = H[\boldsymbol{x}]$$

- Information needed to describe $\boldsymbol{x}$ and $\boldsymbol{y}$ is sum of information about $\boldsymbol{x}$ plus remaining information about $\boldsymbol{y}$ given $\boldsymbol{x}$.

Information Theory

- Kullback-Leibler divergence (Relative entropy): measures the similarity of two distributions
- Approximating distribution q & unknown true distribution p

$$KL(p \parallel q) = -\int p(\boldsymbol{x}) \ln q(\boldsymbol{x})\, d\boldsymbol{x} + \int p(\boldsymbol{x}) \ln p(\boldsymbol{x})\, d\boldsymbol{x}$$

$$= -\int p(\boldsymbol{x}) \ln \frac{q(\boldsymbol{x})}{p(\boldsymbol{x})}\, d\boldsymbol{x}$$

- Not a true measure as it is not symmetric
- $KL(p \parallel q) \geq 0, KL(p \parallel q) = 0 \Leftrightarrow p = q$

Information Theory

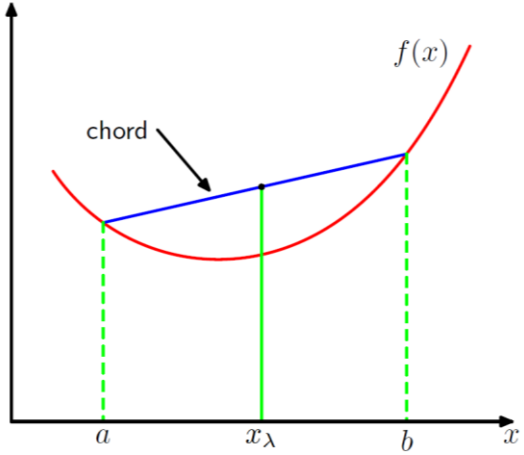
- Review of convex function:

$$x \in [a, b]\ x = \lambda a + (1 - \lambda) b$$

$$f(\lambda a + (1 - \lambda) b) \leq \lambda f(a) + (1 - \lambda) f(b)$$

$$f\left(\sum_{i=1}^{M} \lambda_i x_i\right) \leq \sum_{i=1}^{M} \lambda_i f(x_i)$$

$$\lambda_i > 0, \sum_i \lambda_i = 1$$



Information Theory

- *Jensen's inequality*: considering λ_i as probability distribution over x

$$f(\mathbb{E}[x]) \leq \mathbb{E}[f(x)]$$

$$f\left(\int \boldsymbol{x} p(\boldsymbol{x}) d\boldsymbol{x}\right) \leq \int f(\boldsymbol{x}) p(\boldsymbol{x}) d\boldsymbol{x}$$

Information Theory

- Applying Jensen's inequality to K-L divergence as $-\ln(x)$ is a convex function proves that

$$-\ln\left(\int \frac{q(\boldsymbol{x})}{p(\boldsymbol{x})} p(\boldsymbol{x}) d\boldsymbol{x}\right) \leq -\int p(\boldsymbol{x}) \ln \frac{q(\boldsymbol{x})}{p(\boldsymbol{x})} d\boldsymbol{x}$$

$$-\int p(\boldsymbol{x}) \ln \frac{q(\boldsymbol{x})}{p(\boldsymbol{x})} d\boldsymbol{x} \geq -\ln\left(\int q(\boldsymbol{x}) d\boldsymbol{x}\right) = 0$$

Information Theory

- *Density estimation* is a problem of modeling unknown probability distribution
- Approximating distribution $q(\boldsymbol{x}|\boldsymbol{\theta})$ by minimizing K-L divergence wrt. unknown $p(\boldsymbol{x})$ represented by $\boldsymbol{x}_n$

$$KL(p \parallel q) \simeq \sum_{i=1}^{N} \{-\ln q(\boldsymbol{x}_n|\boldsymbol{\theta}) + \ln p(\boldsymbol{x}_n)\}$$

- Minimizing KL is equivalent to maximum likelihood estimation as it is in form of negative log likelihood after the second term, independent of $\boldsymbol{\theta}$ disappears

Information Theory

- *Mutual information* is degree of correlation of two distributions of random variables $\boldsymbol{x}$ and $\boldsymbol{y}$
- If $p(\boldsymbol{x}, \boldsymbol{y}) = p(\boldsymbol{x})p(\boldsymbol{y})$ variables are independent

$$I[\boldsymbol{x}, \boldsymbol{y}] = KL\big(p(\boldsymbol{x}, \boldsymbol{y}) \parallel p(\boldsymbol{x})p(\boldsymbol{y})\big)$$
$$= -\int\int p(\boldsymbol{x}, \boldsymbol{y}) \ln\left(\frac{p(\boldsymbol{x})p(\boldsymbol{y})}{p(\boldsymbol{x}, \boldsymbol{y})}\right) d\boldsymbol{x}d\boldsymbol{y}$$

Information Theory

- Mutual information is nonnegative and $I[\boldsymbol{x}, \boldsymbol{y}] = 0$ iff $\boldsymbol{x}$ and $\boldsymbol{y}$ are independent
- Represents reduction in uncertainty about $\boldsymbol{x}$ after new observation $\boldsymbol{y}$

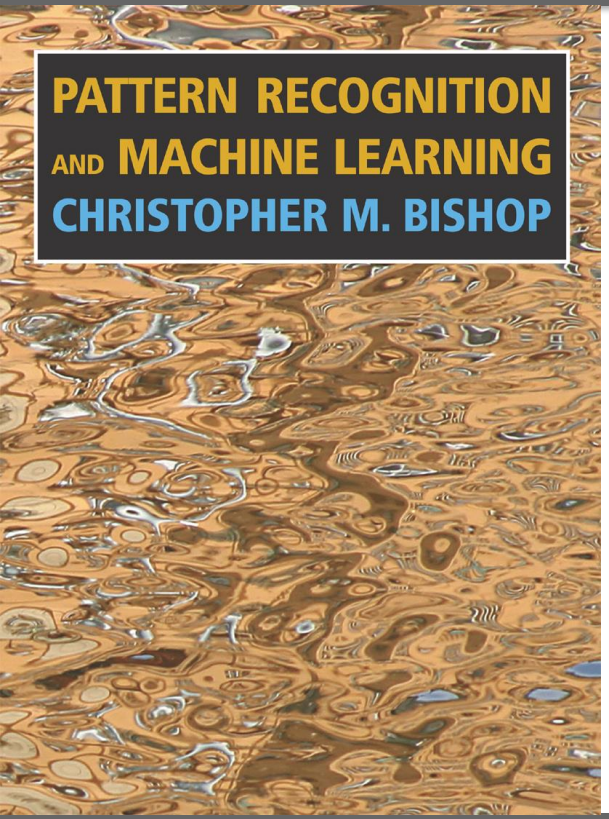
$$I[\boldsymbol{x}, \boldsymbol{y}]$$

$$= -\int\int p(\boldsymbol{x}, \boldsymbol{y}) \ln p(\boldsymbol{x})\, d\boldsymbol{x} d\boldsymbol{y} - \int\int p(\boldsymbol{x}, \boldsymbol{y}) \ln p(\boldsymbol{y})\, d\boldsymbol{x} d\boldsymbol{y}$$

$$+ \int\int p(\boldsymbol{x}, \boldsymbol{y}) \ln p(\boldsymbol{x}, \boldsymbol{y})\, d\boldsymbol{x} d\boldsymbol{y} = H[\boldsymbol{x}] + H[\boldsymbol{y}] - H[\boldsymbol{x}, \boldsymbol{y}] = H[\boldsymbol{x}] - H[\boldsymbol{x}|\boldsymbol{y}]$$

- From Bayesian perspective $p(\boldsymbol{x})$ is prior distribution and $p(\boldsymbol{x}|\boldsymbol{y})$ is a posterior distribution after observing new data $\boldsymbol{y}$

Credits



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